Cyclicity Six and Saturation Geometry at Cancelling Double Grazing

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Abstract

We determine the sharp cyclicity of a semistable limit cycle with two separated quadratic grazings in a continuous piecewise smooth planar system. The active sides agree under transverse transport, two nonzero passage strengths cancel, and the smooth closing curvature is nonzero. The reference displacement is C^2 with an ordinary double zero, yet the cyclicity is six. Changes in the strength sum and activation thresholds prevent joint C^2 regularity of the unfolding.

An exact curvature-zero graph and threshold estimates give this upper bound in every fixed admissible family preserving the prescribed grazing geometry, at arbitrary relative rates of four controls: closing offset, multiplier defect, strength sum and threshold separation. The bounds are four when the strength sum vanishes or the thresholds align, and two when both hold. Full-rank unfoldings attain all four bounds. Six-cycle configurations force weighted bounds on the controls and cycle locations. For positive first passage strength and negative closing curvature, the closures of the rescaled six-cycle domains in a fixed full-rank unfolding converge in Hausdorff distance to a quartic coefficient region with four nonnegative roots, counted with multiplicity.

In this sign case, a fourfold boundary root continues to a cycle of multiplicity four for every small positive strength sum. With two simple companion cycles, its local unfolding gives complete counts of two, four or six simple cycles on the common return interval, off the multiple-root set. A predator–prey model with continuous threshold harvesting realizes the four controls through its natural parameters.

Keywords. Continuous piecewise smooth planar systems; grazing bifurcation; cyclicity; Poincaré return maps; multiple limit cycles.

Mathematics Subject Classification. Primary 34C23; Secondary 34A36, 34C25, 37G15.

1 Introduction and main results

In a jointly C^2 unfolding, a periodic orbit whose return displacement has an ordinary double zero has at most two nearby cycles: the second derivative keeps its sign, and Rolle's theorem applies [31]. We study a continuous piecewise smooth planar system in which the reference displacement has such a double zero, but the sharp local cyclicity is six.

The reference orbit has two spatially separated quadratic grazings, with both active sides agreeing after transport to one transverse section. Each passage contributes a term of order $q^{3/2}$ in a transported transverse coordinate q [6, 22], and their nonzero leading coefficients cancel. The quadratic defect of the composed passages cancels as well, as the expansion below shows. The reference displacement is therefore C^2, with its nonzero quadratic term supplied by the smooth closing map. Under perturbation, the strength sum can become nonzero and the activation thresholds can separate, so the family need not be jointly C^2. Together with

the closing offset and multiplier defect, these give four scalar return-map controls. We answer three questions.

(Q1) What is the cyclicity of the reference cycle?

(Q2) What parameter scales are necessary for six limit cycles, and what is the limiting shape of the six-cycle parameter set?

(Q3) Does a fourfold root of the limiting polynomial continue to a cycle of multiplicity four, and what is its local unfolding?

Theorems 1.2, 1.3 and 1.4 below answer these questions.

1.1 The return map

In each of the two grazing boxes, the planar equation has the local form

$$\dot{\mathbf{x}} = X_\lambda^-(\mathbf{x}) + (h_\lambda(\mathbf{x}))_+ Z_\lambda(\mathbf{x}), \qquad \mathbf{x} \in \mathbb{R}^2, \qquad u_+ = \max\{u, 0\}.$$

The local data are jointly smooth in state and parameters and may differ between the boxes. The active branch $X_\lambda^+ = X_\lambda^- + h_\lambda Z_\lambda$ agrees with X_λ^- on the seam $\{h_\lambda = 0\}$, so the field is continuous there. The boxes are joined by regular flow arcs; Definition 2.1 gives the precise admissibility conditions. The threshold-harvesting family of Theorem 6.3 realizes these conditions.

In a transverse coordinate q constructed in Section 2, the return map and the displacement have the form

$$P(q) = R\{\delta + \Gamma_2(\Gamma_1(q) - \delta)\}, \qquad D(q) = P(q) - q.$$

Here Γ_1, Γ_2 are the passage maps near the two tangencies, R is the smooth transition along the rest of the cycle, and parameter dependence is suppressed. The second passage is activated by the intermediate input $\Gamma_1(q) - \delta$, so the two grazing thresholds on the section are 0 and $\Gamma_1^{-1}(\delta)$. Section 2 derives this composition from the vector field and shows that it is the first return on a fixed interval.

We use four return-map controls,

$$a = R(0), \qquad b = R'(0) - 1, \qquad m = A_1 + A_2, \qquad \delta,$$

namely the offset, the multiplier defect, the strength sum and the threshold separation. They are smooth functions of the physical parameters and need not be independent. All four vanish at the reference parameter, where $A_1 = -A_2 \neq 0$ and $c_0 = R''(0)/2 \neq 0$. Writing $q_+ = \max\{q, 0\}$, the correction $\Gamma_\nu(q) - q$ begins with $A_\nu q_+^{3/2}$, and the reversibility of the quadratic passage fixes its next coefficient at $3A_\nu^2/4$ (Proposition 2.4). Consequently, when the thresholds are aligned,

$$\Gamma_2(\Gamma_1(q)) - q = m q_+^{3/2} + \tfrac{3}{4} m^2 q_+^2 + O(q_+^{5/2}) \qquad (\delta = 0).$$

Both displayed terms vanish when $m = 0$, so the quadratic term of the reference displacement comes from R (Lemma 2.5). For $\delta \neq 0$, the two $q^{3/2}$ contributions start at different thresholds. Their singular curvatures have opposite signs and compete. Neither an outer expansion nor the number of active intervals controls the fixed-point count on the whole return interval; estimates relative to the moving thresholds are needed.

We use the standard notion of cyclicity [31].

Definition 1.1 (Cyclicity)**.** Let (X_λ) be a family of planar vector fields and let $\mathcal{C}$ be a periodic orbit of X_{λ_0}. Let $N(\lambda, \varepsilon)$ be the number of limit cycles of X_λ at Hausdorff distance less than ε from $\mathcal{C}$. The cyclicity of $\mathcal{C}$ in (X_λ) is

$$\mathrm{Cycl}\big((X_\lambda), \mathcal{C}\big) = \inf_{\varepsilon,\eta>0} \sup_{|\lambda-\lambda_0|<\eta} N(\lambda, \varepsilon).$$

For a class $\mathfrak{F}$ of families through X_{λ_0}, $\mathrm{Cycl}(\mathfrak{F}, \mathcal{C})$ denotes the supremum over the families in $\mathfrak{F}$.

The families we consider are the admissible families of Definition 2.1: finite-dimensional smooth families of continuously matched fields that preserve regular switching curves, the two quadratic tangencies, their transported active side and the one-turn itinerary of $\mathcal{C}$, with $a = b = \delta = 0$, $A_1 = -A_2 \neq 0$ and $c_0 \neq 0$ at the reference parameter. When the map $\lambda \mapsto (a, b, m, \delta)$ is a submersion at λ_0, we fix a local inverse

$$\Lambda(a, b, m, \delta) = \lambda(a, b, m, \delta; \xi_0),$$

with the remaining parameters fixed at ξ_0, and call the resulting fixed four-parameter family the *standard unfolding*.

1.2 Sharp cyclicity

Theorem 1.2 (Cyclicity)**.** *Let $\mathfrak{F}$ be the class of admissible families through the reference field.*

1. *For every family in $\mathfrak{F}$ there are a neighbourhood U of λ_0 and an interval $I = (-q_0, q_0)$ on the transverse section such that, for every $\lambda \in U$, the return map P_λ has at most six fixed points in I; at most four if $m(\lambda) = 0$ or $\delta(\lambda) = 0$; and at most two if $m(\lambda) = \delta(\lambda) = 0$.*

2. *If $\lambda \mapsto (a, b, m, \delta)$ is a submersion at λ_0, all four bounds are attained by hyperbolic fixed points along smooth parameter paths in the corresponding slices that tend to λ_0. For every sufficiently small m with $mc_0 < 0$, the standard unfolding has six hyperbolic fixed points of order m^2 in I whenever $(a/m^4, b/m^2, \delta/m^3)$ belongs to a fixed nonempty open set. Three cycles are attracting and three repelling.*

3. *Consequently $\mathrm{Cycl}(\mathfrak{F}, \mathcal{C}) = 6$. The corresponding suprema over admissible families contained in $\{m = 0\}$, in $\{\delta = 0\}$ and in $\{m = \delta = 0\}$ are four, four and two.*

The proof controls curvature before counting fixed points. With the other controls fixed, the outer equation $P''(q) = 0$ is represented exactly by a separation graph $\delta = \Delta(r)$, $r = \sqrt{q}$. This graph has at most one positive critical point, and hence at most two intersections with a fixed separation level. Estimates relative to the moving thresholds exclude further curvature zeros outside the outer region. The two possible curvature zeros and the two thresholds divide the continuous derivative P' into at most five strictly monotone pieces. Thus $P' - 1$ has at most five zeros, and Rolle's theorem gives at most six fixed points, including those at the thresholds.

Theorem 1.2 is proved in Section 3 (Proposition 3.4, Corollary 3.5 and Propositions 3.9–3.10) and Section 6.1. Proposition 6.1 constructs, through every reference field, an admissible family for which $\lambda \mapsto (a, b, m, \delta)$ is a submersion; Corollary 6.2 gives the same cyclicity in the smaller class of local perturbations used there.

1.3 The saturation domain

For an admissible family we call the set of parameters $\lambda \in U$ for which P_λ has six fixed points in I the *saturation domain*, and a sequence of such parameters tending to λ_0 a *saturated sequence*. Describing this set requires estimates that remain valid when the parameters tend to

the reference at arbitrary relative rates, including configurations whose fixed points approach the thresholds. Six zeros of D force five alternating extrema, and their locations and values constrain the parameters.

Consider the sign case $A_1(\lambda_0) = B > 0$, $c_0 = -C < 0$. Saturation forces $m > 0$, $\delta > 0$ and $b < 0$. Beyond both thresholds, the balance of $mq^{3/2}$ and $-Cq^2$ suggests q of order m^2. At this scale, bq, the separation term $\frac{3}{2}B\delta\sqrt{q}$ and the offset a have the same order when b, δ, a have weights $2, 3, 4$ relative to m. Theorem 1.3 shows that every six-cycle configuration obeys upper bounds with these weights.

Define the normalized parameters and the normalized positive state by

$$A_4 = \frac{C^3 a}{m^4}, \qquad S_4 = -\frac{Cb}{m^2}, \qquad T_4 = \frac{3BC^2\delta}{2m^3}, \qquad x = \frac{C\sqrt{q}}{m}.$$

The limiting polynomial is

$$\Phi_{A,S,T}(x) = A - Sx^2 + x^3 - x^4 + Tx.$$

With e_j denoting the elementary symmetric polynomials of $r = (r_1, r_2, r_3, r_4)$, set

$$\mathcal{K}_4 = \left\{ (-e_4(r), e_2(r), e_3(r)) : r_i \geq 0,\ \sum_{i=1}^{4} r_i = 1 \right\}.$$

This is the compact set of coefficients for which $\Phi_{A,S,T}$ has four nonnegative real roots, counted with multiplicity. For the standard unfolding and $m > 0$, let $\mathscr{T}_m$ be the set of (A, S, T) in a fixed compact box $\mathcal{B}$ for which the parameters $(a, b, m, \delta) = (m^4 A/C^3,\ -m^2 S/C,\ m,\ 2m^3 T/(3BC^2))$ give six fixed points in I; the box $\mathcal{B}$ contains all normalized six-cycle parameters (see (4.10)). We write d_H for the Hausdorff distance in $\mathbb{R}^3$.

Theorem 1.3 (Saturation at the principal scale)**.** *1. For every admissible family, after shrinking U and I if necessary, there is $K > 0$ such that, whenever $\lambda \in U$ and P_λ has six fixed points $q_1, \ldots, q_6$ in I,*

$$0 < |\delta| \leq K|m|^3, \quad |b| \leq K|m|^2, \quad |a| \leq K|m|^4, \quad \max_j |q_j| \leq K|m|^2.$$

2. For every admissible family in the sign case $A_1(\lambda_0) = B > 0$, $c_0 = -C < 0$, every accumulation point of (A_4, S_4, T_4) along a saturated sequence belongs to $\mathcal{K}_4$.

3. In the same sign case, for the standard unfolding,

$$d_H(\overline{\mathscr{T}_m}, \mathcal{K}_4) \longrightarrow 0 \quad (m \to 0^+).$$

For every compact set $E \subset \operatorname{int} \mathcal{K}_4$ there is $m_E > 0$ such that $E \subset \mathscr{T}_m$ for $0 < m < m_E$, and the six fixed points are simple.

For $0 < m < m_E$, a point of E gives four fixed points beyond both thresholds and two with $q < 0$, where both passages are inactive. Boundary points of $\mathcal{K}_4$ correspond to zero or repeated roots of $\Phi_{A,S,T}$. Theorem 1.4 continues the fourfold limiting root to a nearby parameter branch with one cycle of multiplicity four and two simple cycles. Theorem 1.3 is proved in Section 4: part 1 is Proposition 4.3, part 2 is Proposition 4.6, and part 3 is completed at the end of Section 4.1.

1.4 Limit cycles of multiplicity four

The boundary point

$$(A, S, T) = \left(-\frac{1}{256}, \frac{3}{8}, \frac{1}{16}\right)$$

has four roots equal to $1/4$, so $\Phi_{A,S,T}(x) = -(x - 1/4)^4$. The next theorem continues this configuration to the vector field for every small positive m and counts all nearby limit cycles.

Theorem 1.4 (Limit cycles of multiplicity four)**.** *Consider the standard unfolding in the sign case $A_1(\lambda_0) = B > 0$, $c_0 = -C < 0$. For every sufficiently small $m > 0$ there are parameters $(a_*(m), b_*(m), \delta_*(m))$ at which P_λ has a fixed point of multiplicity four beyond both thresholds and two simple fixed points with $q < 0$. Near these parameters, with m fixed, smooth changes of the state and of (a, b, δ) reduce the fixed-point equation near the fourfold fixed point, up to a nonvanishing factor, to*

$$y^4 + uy^2 + vy + w = 0.$$

Off the set where this quartic has a multiple root, P_λ has two, four or six fixed points in I, and each count occurs on a nonempty open set of parameters. At the fourfold point P_λ has exactly three fixed points in I.

The proof first locates all zeros of D in I (Lemma 5.2): two simple zeros with $q < 0$ and a cluster beyond both thresholds, at distance of order m^2 from them. On this cluster D is smooth in q, so the Malgrange preparation theorem applies once the fourfold zero has been constructed in normalized variables (Theorem 5.3). The exact identities $q_* = r_c^2$ and $\delta_* = \delta_c$ in that theorem place the constructed cycle at the critical point of the separation graph. The fold surface, the curve of triple roots and the curve of two double roots are then explicit (Proposition 5.4). Theorem 1.4 follows from Lemma 5.2 and Theorem 5.3.

1.5 Realization and further results

Section 6 shows that the hypotheses hold in two ways: for local perturbations of an arbitrary reference field (Proposition 6.1), and for the natural parameters of a predator–prey model with continuous threshold harvesting (Theorem 6.3). The model combines the smooth core of Huang, Gong and Ruan [12] with continuous threshold policies [1, 16, 29, 27]. In this model the reference cycle, the cancellation, the negative closing curvature and the submersion condition are all obtained with positive ecological parameters.

Online Resource 1 treats the corner $S_4 \to 0$ of $\mathcal{K}_4$, where three roots tend to zero, at a secondary scale. It identifies the limiting cubic coefficient region, constructs six limit cycles with positive offset $a > 0$, and unfolds a cusp at the triple root. The proofs in the present paper do not use these results.

1.6 Relation to previous work

The $q^{3/2}$ correction for continuous fields and the composition of successive grazing maps belong to the established theory of grazing bifurcations [21, 6, 22, 7, 5]. For a periodically forced oscillator with a continuous piecewise-linear restoring force, Liu and Yue compose two such passages with regular transitions and test the second activation at the intermediate state [19]. For bilateral impacts, Li, Wei, Tan and Xi compose two local impact maps and give a necessary existence condition [17, Theorem 1]; related compositions occur in [18, 20]. Two-parameter grazing bifurcations, including simultaneous contacts at separated points, are classified in [13]. We take the composition rule as known and determine the cyclicity of the resulting return map.

Translated-power compositions, studied by Dumortier and Roussarie [8], provide a related approach to counting return-map zeros. Panazzolo's generalized derivation–division algorithm gives root bounds for these compositions and parameter-analytic compensator expansions [28, arXiv version, Theorem 2.2, Sections 3–4 and Proposition 5.2]. That explicit function class has not been matched here to the actual return, with its smooth profiles, remainders and closing map, across moving activation thresholds. We count its curvature zeros through the exact separation graph and uniform threshold estimates.

Bifurcations of periodic orbits in planar piecewise smooth systems have been classified in low codimension [15, 11], and limit cycles of continuous piecewise linear systems are studied in [9]; see also [32]. A related configuration is a periodic orbit tangent to two switching lines in a discontinuous system with sliding, whose bifurcations resemble those of double and triple cycles [30]. Here the field is continuous, there is no sliding, and we prove cyclicity six for a semistable cycle with cancelling leading passage coefficients.

Nonhyperbolic cycles and cancellations also occur in other grazing problems. For one continuous planar grazing contact, Simpson and Meiss obtain a return with a $q^{3/2}$ term, a nearby fold and a sixth-order separation of the fold and grazing curves [34, Theorem 1]. Colombo and Dercole [4] study discontinuity-induced bifurcations of nonhyperbolic cycles, including a fold with multiplier one. In resonant impact grazing, a singular transport contribution is suppressed while the local square-root correction remains nonzero [33, Theorems 2.4–2.5]. T. Matsuhisa (*Induced-contact stability and activation thresholds near double compliant grazing*, preprint dated 3 October 2026, SSRN:7556610) studies height-resolved corrections and paths cancelling the direct second-gap height against the transmitted first-contact height. The continuation of the matched periodic branch uses the assumption that $D_z P_f - I$ is invertible (Theorem 2). Here two nonzero transported passage strengths cancel at a reference cycle with multiplier one, so this continuation argument is unavailable.

Large numbers of limit cycles at discontinuous tangencies arise from a different local geometry. Novaes, Teixeira and Zeli study a cycle through a visible two-fold point [25]. Novaes and Silva construct five hyperbolic limit cycles near one monodromic tangential singularity of a quadratic Filippov family, with a sixth created by a pseudo-Hopf perturbation [23, Example 3 and Remark 1]; further constructions appear in [24]. These tangencies occur at a single point of a Filippov system, whereas the two contacts here are separated and the fields are continuously matched. For Sotomayor–Teixeira regularization, Bonet-Revés and M-Seara treat an identity-return centre and hyperbolic grazing cycles [2, Theorems 2.3–2.4], and prove a unique saddle-node for repelling hyperbolic grazing [3, Theorems 2.4 and 2.10].

Two preprints of the author treat related nonhyperbolic returns: an invariant-line entry–exit followed by a separated quadratic grazing (H. Lu, *Neutral Entry–Exit Cycles with Quadratic Grazing: Uniform Return Reduction and Local Two-Parameter Bifurcations*, preprint, 2026, arXiv:2607.27464v4), and a single high-order grazing with multiwell passages (H. Lu, *Neutral Returns at High-Order Grazing: Sharp Cyclicity, Physical Codimension, and Weighted Crossover*, preprint, 2026, arXiv:2608.01447v3). In the high-order case the active curvature contributions have the same sign and reinforce each other; here they have opposite signs and compete.

1.7 Organization

Section 2 constructs the return map, derives the passage law and proves the estimates used throughout. Section 3 proves the bounds of Theorem 1.2 and their attainment. Section 4 proves Theorem 1.3. Section 5 constructs the limit cycles of multiplicity four and proves Theorem 1.4. Section 6 realizes the hypotheses by local perturbations and in a harvesting model, and Section 7 lists open problems. Appendix A constructs the smooth-core fold used in the harvesting model.

2 The return map and uniform estimates

The two contact boxes and their regular connections determine a return map on one transverse section. We first establish its itinerary and the reversible passage law, then derive the cancellation and the value estimates used in the fixed-point count.

Conventions. A family is fixed throughout. All domains and bounds are chosen on compact reductions that are common to the whole family and do not depend on how the parameters approach the reference; derivative orders are prescribed in advance. Derivatives in the state are taken at fixed parameters, and derivatives and multiplicities refer to the transverse coordinate q. A function of $\sqrt{q}$ that extends smoothly through $\sqrt{q} = 0$ is called a *root profile*; smoothness of a root profile does not imply smoothness in q. The letters K and C_j denote constants that may change from line to line, and letters for smooth remainders, such as L, M, N, S, E, are local to the statement or proof in which they appear.

2.1 Geometry and a common first return

Definition 2.1 (Admissible family)**.** A finite-dimensional smooth family of continuously matched planar fields, parametrized by λ, is *admissible* if the following hold near its reference parameter λ_0.

1. The branch fields, switching curves, marked contacts and transverse sections are jointly smooth. In each of two disjoint contact boxes,

$$X_\lambda^+ = X_\lambda^- \quad \text{on } \{h_\lambda = 0\}, \qquad X_\lambda^+ - X_\lambda^- = h_\lambda Z_\lambda. \tag{2.1}$$

   The switching function h_λ has nonzero differential, the inactive field X_λ^- does not vanish, and its marked orbit has a persistent quadratic contact: at each marked contact,

$$X_\lambda^- h_\lambda = 0, \qquad (X_\lambda^-)^2 h_\lambda \le -\kappa_0 < 0$$

   for one common $\kappa_0 > 0$. The active region is $\{h_\lambda > 0\}$, where X_λ^+ is used; near each marked contact the marked orbit lies in $\{h_\lambda \le 0\}$.

2. Each contact box has an incoming and an outgoing transverse section (its ports), with closures in the interior of the box, and consecutive boxes are joined by orbit arcs of X_λ^- along which $h_\lambda < 0$. On a common neighbourhood of states and parameters, an orbit that enters a box with positive penetration (the local input z defined below) crosses the seam into the active region, stays there, and leaves through the seam before reaching the outgoing port; an orbit with nonpositive penetration touches the seam at most once and otherwise stays in $\{h_\lambda < 0\}$. On one fixed incoming interval $I = (-q_0, q_0)$, including its grazing thresholds, every orbit follows this itinerary, meets no seam along the connecting arcs, and does not return to the incoming section before completing one turn. All intermediate port images lie in common open domains. The local penetration coordinates are positive towards the active regions. The regular transition from the first outgoing port to the second incoming port is increasing in these coordinates, with a common positive derivative margin. Thus both active sides agree after transport to the incoming section.

3. At λ_0 there is a compact embedded nonstationary periodic orbit $\mathcal{C}$ through the two marked contacts. Its return-map controls, passage strengths and closing curvature, defined in (2.9) and (2.10) below, satisfy

$$a = b = \delta = 0, \qquad A_1 = -A_2 \neq 0, \qquad c_0 \neq 0.$$

We transport the inactive first integrals along the regular connections to one smooth transverse coordinate q, with contact levels $\tau_1 = 0$ and $\tau_2 = \delta$; Proposition 2.6 below shows that this is possible for nonidentity connecting maps. In the νth box put $z = q - \tau_\nu$. An inactive flow box gives the seam $z = s^2\gamma(s,\lambda)$ with $\gamma > 0$; the signed clock coordinate $t = s\sqrt{\gamma(s,\lambda)}$ leaves z unchanged and normalizes the seam. After a positive time change, matching and smooth division give

$$X^- = \partial_t, \qquad \Sigma = \{z = t^2\}, \qquad X^+ = (1 + h\alpha)\partial_t + h\mu\partial_z, \qquad h = z - t^2. \tag{2.2}$$

The functions α, μ are jointly smooth and the active clock is positive on a common smaller box. This normalization preserves oriented orbits and section maps, so it does not change the return map. Figure 1 shows the reference itinerary and the grazing thresholds on the incoming section.

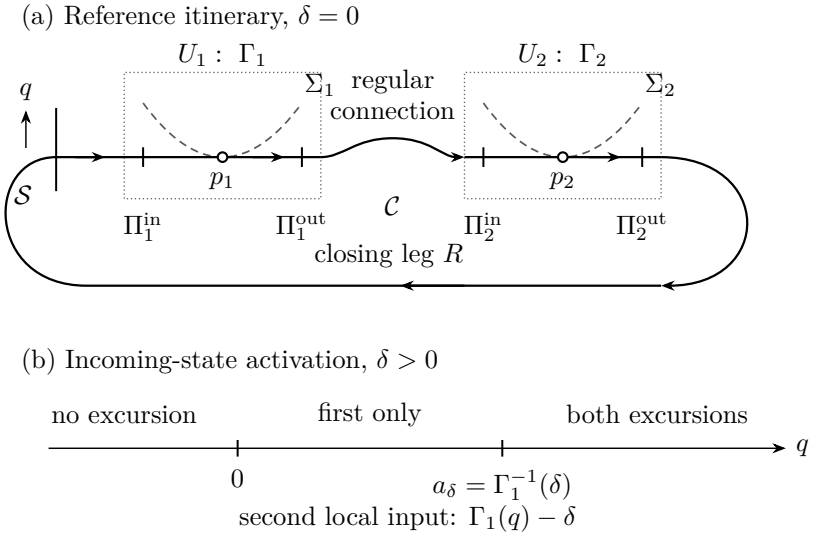


Figure 1: Schematic return geometry. Top: the reference itinerary at $\delta = 0$, with dotted contact boxes U_1, U_2, dashed switching seams Σ_1, Σ_2, transverse ports and the return section $\mathcal{S}$. Arrows on the cycle show the direction of time; the section arrow orients q. Bottom: activation on the incoming state axis for $\delta > 0$. The second passage is active when $\Gamma_1(q) - \delta > 0$, so its threshold is a_δ. For $\delta < 0$, $a_\delta = \delta < 0$ lies to the left of 0, and only the second excursion occurs between them.

Lemma 2.2 (A common one-turn return tube)**.** *Let a compact embedded nonstationary reference cycle admit fixed tubular coordinates (θ, w), with $\theta \in \mathbb{R}/\mathbb{Z}$, reference $w = 0$ and section $\theta = 0$. Suppose a finite smooth family of matched fields has common local Lipschitz bounds on the tube and differs from the reference field at λ_0 by $O(|\lambda - \lambda_0|)$. Fix contact cores and transverse ports with strict margins, and strictly inactive connecting arcs. In every core, assume that orbits with positive penetration enter the active region, stay there and leave through the seam inside the core, and that the other orbits touch the seam at most once. If the angular clock remains positive and these local and port margins persist, one can choose a common input interval and a parameter neighbourhood on which the chronological composition is the first return map. It makes one whole turn, with no further seam crossing or earlier section hit.*

Proof. The matched form $X_\lambda^- + (h_\lambda)_+ Z_\lambda$ has common Lipschitz bounds. Compactness gives $\dot\theta \geq v_* > 0$ on a smaller tube; division by this clock and Gronwall comparison with $w = 0$ give,

up to a first exit,

$$\left|\frac{dw}{d\theta}\right| \le L|w| + K|\lambda - \lambda_0|,$$
$$|w(\theta)| \le e^L(|w(0)| + K|\lambda - \lambda_0|), \quad 0 \le \theta \le 1.$$

Choose one input interval and parameter neighbourhood keeping this bound strictly inside the tube, core and port margins. First-exit continuation gives a full turn; the assumed core passages and inactive arcs exclude further seam crossings, including for tangent inputs. The ports join on common domains, and strict angular advance makes $\theta = 1$ the next section hit. The return image need not lie in the input interval. □

Lemma 2.2 gives a geometric criterion for the first-return itinerary in item 2 of Definition 2.1; Section 6 applies it to the constructed families.

Remark 2.3 (Cycles and fixed points). After a common reduction, every periodic orbit sufficiently close to $\mathcal{C}$ in the return tube meets the transverse section in I. The first return is increasing and has no nontrivial finite-period orbit whose iterates remain in the common local domain: if $P(q) > q$ or $P(q) < q$, its successive iterates are strictly ordered. Thus these periodic orbits correspond to fixed points in I, and isolation of the fixed point is equivalent to isolation of the cycle in this tube.

2.2 Reversible passages and cancellation

An active leaf pairs its two seam intersections by a decreasing involution. We use it to identify the entry and exit of the passage. The resulting relation between its first two coefficients determines cancellation in the two-passage composition.

Proposition 2.4 (Passage map near a quadratic grazing)**.** *For a local passage with the quadratic geometry* (2.2) *and fixed local transverse ports, there is a jointly smooth signed companion $g(t, \lambda)$ on a two-sided interval such that*

$$g(0, \lambda) = 0, \qquad g_t(0, \lambda) = 1, \qquad \iota g \iota = g^{-1}, \qquad \iota(t) = -t. \tag{2.3}$$

The passage in the inactive coordinate is

$$\Gamma(q, \lambda) = q \quad (q \le 0), \qquad \Gamma(r^2, \lambda) = g(r, \lambda)^2 \quad (r > 0). \tag{2.4}$$

Its first entry is $t = -r$ and its first moving exit is $g(r, \lambda)$. On a common domain,

$$\Gamma(q, \lambda) = q + C_1(\lambda) q_+^{3/2} + \frac{3}{4} C_1(\lambda)^2 q_+^2 + q_+^{5/2} L(\sqrt{q_+}, \lambda), \tag{2.5}$$

where L is the restriction of a jointly smooth function, and

$$C_1(\lambda) = \frac{4}{3}\mu(0, 0, \lambda). \tag{2.6}$$

The passage is an orientation-preserving C^1 local diffeomorphism, and the companion, its inverse and the root profiles satisfy uniform bounds of every prescribed order.

Proof. Divide the active field by its positive t component. Let $K(t, z, \lambda)$ be the z coordinate where its orbit meets $t = 0$. Smooth finite-time ODE dependence gives $K(0, z, \lambda) = z$ and $K_z > 0$ on a common smaller box. Since the active orbit equation has zero z component on the seam, $K_t(t, t^2, \lambda) = 0$. Hence the seam restriction $\widehat{K}(t, \lambda) = K(t, t^2, \lambda)$ satisfies

$$\widehat{K}_t = 2tE(t, \lambda), \qquad E(t, \lambda) = K_z(t, t^2, \lambda) > 0,$$
$$\widehat{K} = t^2 M(t, \lambda), \qquad M(t, \lambda) = 2\int_0^1 sE(st, \lambda)\, ds.$$

Here $E(0,\lambda) = M(0,\lambda) = 1$, so $\mathcal{N}(t,\lambda) = t\sqrt{M(t,\lambda)}$ is a smooth increasing local diffeomorphism. The active leaf pairs its seam intersections by

$$J = \mathcal{N}^{-1}\iota\mathcal{N}, \qquad J(0) = 0, \quad J'(0) = -1, \quad J^2 = \mathrm{id}.$$

For input $q = r^2 > 0$, the inactive orbit first reaches the seam at $-r$. Its active leaf has value $\widehat{K}(-r) > 0$ and second intersection $e = J(-r) > 0$. The seam restriction decreases for $t < 0$ and increases for $t > 0$; since $K_z > 0$, the leaf $K = \widehat{K}(-r)$ lies strictly above the seam precisely between $-r$ and e. The positive clock makes them the first entry and first exit. In the normalized clock, $dh/dt = -2t$ on the seam, so the entry and exit derivatives are exactly $2r$ and $-2e$. In the original physical time they are multiplied by the positive clock factor. After exit the inactive orbit has $z = e^2$ and lies below the seam; port localization excludes further seam crossings. Inputs $q < 0$ stay below it and $q = 0$ only touches it, giving the identity passage on $q \le 0$.

Set $g = J\iota$. Then $J^2 = \iota^2 = \mathrm{id}$ gives $\iota g \iota = g^{-1}$ and $g'(0) = 1$, while $e = g(r)$ proves (2.4). For negative t, reversal gives

$$g(t,\lambda) = -\sqrt{\Gamma^{-1}(t^2,\lambda)} \qquad (t < 0). \tag{2.7}$$

This signed continuation records backward active travel; the forward passage on negative q is the identity.

Writing $g(t) = t + \rho t^2 + \sigma t^3 + O(t^4)$ and comparing cubic coefficients in $(g\iota)^2 = \mathrm{id}$ gives $\sigma = \rho^2$, the classical jet relation for reversible maps [26, 23]. Taylor's integral remainder therefore gives

$$g(t) = t + \rho t^2 + \rho^2 t^3 + t^4 G(t,\lambda)$$

with G jointly smooth. Squaring at $t = \sqrt{q} > 0$ gives $C_1 = 2\rho$ and $C_2 = 3\rho^2 = 3C_1^2/4$, proving (2.5). Its parameter derivatives also give joint C^1 regularity and $\Gamma_q \to 1$ at zero.

For (2.6), note that $dz/dt = f(t,z) = h\mu/(1 + h\alpha)$ and $K_t = -fK_z$. On the seam $f = 0$ and $f_z = \mu$, so $E'(0) = -\mu(0,0)$. Thus $\mathcal{N}(t) = t - \mu(0,0)t^2/3 + O(t^3)$, and $\mathcal{N}^{-1}\iota\mathcal{N}\iota$ has quadratic coefficient $2\mu(0,0)/3$; squaring gives (2.6). Compactness, smooth dependence on initial data and a common shrinking give uniform positivity and the prescribed derivative bounds. □

Since $\partial_q = (2r)^{-1}\partial_r$, the jth state derivative of $q^{5/2}L(\sqrt{q},\lambda)$ is bounded by $C_j q^{5/2-j}$ on $q > 0$, for every fixed j and after finitely many parameter derivatives. Its zero extension to $q \le 0$ is jointly C^2 in state and parameters.

Write Γ_1, Γ_2 for the two normalized passages. The inactive connections preserve q, so chronological continuation gives

$$\begin{aligned} Q_{m,\delta}(q,p) &= \delta + \Gamma_2(\Gamma_1(q,p,m,\delta) - \delta, p, m, \delta), \\ P_{m,\delta} &= R_{p,m,\delta} \circ Q_{m,\delta}, \end{aligned} \tag{2.8}$$

where R is the smooth orientation-preserving transition along the rest of the cycle, the *closing map*. Write $D = P - q$ and $A_\nu(\lambda) = C_{1,\nu}(\lambda)$, the *passage strengths*. When the controls are independent we write $p = (a,b,\xi)$ and $\lambda = (a,b,m,\delta,\xi)$, with ξ in a fixed compact set. Throughout, m denotes the strength sum:

$$A_1(p,m,\delta) + A_2(p,m,\delta) = m, \qquad A(p) = A_1(p,0,0). \tag{2.9}$$

The individual strengths and all higher profiles may depend on every parameter. After reduction, A has fixed sign with $|A(p)| \ge A_* > 0$, and the two passage strengths have opposite signs. The jets of the closing map are

$$a = R(0), \qquad b = R'(0) - 1, \qquad c = \tfrac{1}{2}R''(0). \tag{2.10}$$

In particular, the aligned multiplier is $k(p) = R'_{p,0,0}(0) = 1 + b > 0$. Taylor division gives

$$R(y) = a + (1 + b)y + c(\lambda)y^2 + y^3 S(y, \lambda). \tag{2.11}$$

At the reference $a = b = m = \delta = 0$. The aligned curvature $c_0(p) = \frac{1}{2}R''_{p,0,0}(0)$ has fixed nonzero sign and a uniform strict margin; after shrinking, R'' retains that sign at every relevant input. Its value may vary with the parameters. The attainment results use the condition

$$\operatorname{rank} D_\lambda(a, b, m, \delta) = 4 \quad \text{at the reference field,} \tag{2.12}$$

which makes (a, b, m, δ, ξ) local parameter coordinates.

Lemma 2.5 (Cancellation at aligned thresholds)**.** *On $\delta = 0$, the composition has the root-profile form*

$$Q_{m,0}(q) = q + mq_+^{3/2} + \frac{3}{4}m^2 q_+^2 + q_+^{5/2} L_0(\sqrt{q_+}, a, b, m, \xi). \tag{2.13}$$

Consequently, on the whole slice $m = \delta = 0$,

$$Q_{0,0}(q) = q + q_+^{5/2} L_0(\sqrt{q_+}, a, b, 0, \xi) \tag{2.14}$$

is jointly C^2. Only the leading strengths have to cancel.

Proof. On a common smaller domain positive inputs and intermediate images stay positive. The cubic companion jets and $A_1 + A_2 = m$ give

$$g_2 g_1(t) = t + \frac{m}{2}t^2 + \frac{m^2}{4}t^3 + t^4 G_0(t, a, b, m, \xi),$$

with G_0 jointly smooth. Squaring proves (2.13); for $q \le 0$ both passages are identity. At $m = 0$ the remaining $q_+^{5/2}$ profile has the joint C^2 extension proved above. □

The grazing thresholds on the incoming section are

$$0 \quad \text{and} \quad a_\delta = \Gamma_1^{-1}(\delta, \lambda). \tag{2.15}$$

Here a_δ labels the second threshold; it does not denote a parameter derivative. The second local input is $I_2 = \Gamma_1(q, \lambda) - \delta$. The increasing first passage has derivative near one, so its inverse exists on a common interval and

$$a_\delta = \delta \quad (\delta \le 0), \qquad a_\delta = \left[g_1^{-1}(\sqrt{\delta}, \lambda)\right]^2 \quad (\delta > 0). \tag{2.16}$$

Thus a_δ has the sign of δ, $|a_\delta| \le 2|\delta|$, and for positive separation $a_\delta = \delta - C_{1,1}(\lambda)\delta^{3/2} + O(\delta^2)$ with a smooth positive-root profile. It is jointly C^1; as a function of the independent separation variable δ, it is not C^2 at $\delta = 0$ when $C_{1,1} \neq 0$. For $\delta > 0$, the intervals $q < 0$, $0 < q < a_\delta$, $q > a_\delta$ have respectively no excursion, only the first, and both excursions. For $\delta < 0$, the corresponding intervals are $q < \delta$, $\delta < q < 0$, $q > 0$, with only the second excursion in the middle. At either endpoint its excursion becomes a tangency touch. At alignment, negative inputs are inactive, zero touches both contacts and positive inputs make both excursions.

On the aligned cancelling slice, the closing expansion gives

$$D_{0,0}(q, p) = a + bq + c_0(p)q^2 + q_+^{5/2} M(\sqrt{q_+}, p) + q^3 S(q, p), \qquad c_0(p) = \tfrac{1}{2}R''_{p,0,0}(0) \neq 0. \tag{2.17}$$

The root remainder is jointly C^2, with bounds $|\partial_q^j \partial_p^\beta (q^{5/2} M)| \le C_{j,\beta} q^{5/2-j}$ for $j \le 2$. At $a = b = 0$ the reference displacement consequently has $D(0) = D_q(0) = 0$ and $D_{qq}(0) = 2c_0 \neq 0$. The return map is jointly C^1 in state and parameters. Since both strengths are nonzero, P_λ is

C^2 at both thresholds if and only if $m(\lambda) = \delta(\lambda) = 0$; otherwise a nonzero coefficient of order $q^{3/2}$ survives at one of the thresholds. For the counting bounds, fixed points at thresholds are counted as distinct zeros; no smooth multiplicity is used at a non-C^2 threshold.

A smooth increasing change of transverse coordinate $x = \phi(q)$, $\phi(0) = 0$, $\phi'(0) = s > 0$, changes the reference coefficients to $\widetilde{c}_0 = c_0/s$ and $\widetilde{A}_\nu = A_\nu(\lambda_0)/\sqrt{s}$; the terms involving $\phi''(0)$ cancel at a fixed point with unit multiplier. In particular $c_0/A_1(\lambda_0)^2$ does not depend on the coordinate. The odd root lift $t\sqrt{\phi(t^2)/t^2}$ preserves reversal, and nonlinear terms of ϕ first affect the order $5/2$, so the coefficient law of Proposition 2.4 is preserved. These invariants let us check the sign conditions in any such section coordinate and compare the constants in the weighted limits.

2.3 Regular connections and auxiliary parameters

We now show how regular connecting maps lead to this form, and how families with dependent controls are handled.

Proposition 2.6 (Return map through regular connections)**.** *In the common geometry above, let S_1, S_2 be the regular transitions in positive penetration coordinates, with all intermediate images in their common domains. They are smooth and increasing, with common positive derivative margins. If the second local level is δ_2, then*

$$P = S_2 \circ \Pi_2 \circ S_1 \circ \Gamma_1, \qquad \Pi_2(z) = \delta_2 + \Gamma_{2,\mathrm{loc}}(z - \delta_2) \tag{2.18}$$

has the transported form

$$P(q) = R\{\delta + \Gamma_2(\Gamma_1(q) - \delta)\}, \qquad \delta = S_1^{-1}(\delta_2), \quad R = S_2 \circ S_1. \tag{2.19}$$

The second passage retains reversal and the coefficient law. The four controls are smooth functions of λ, and under (2.12) they give a chart (a, b, m, δ, ξ) with $m = C_{1,1} + C_{1,2}$.

Proof. The first transition is increasing by admissibility. The inactive reference monodromy preserves transverse orientation, so $S_2' > 0$ at the reference as well; smooth dependence gives common positive derivative margins after reduction. Set $T(x) = S_1(\delta + x) - \delta_2$ and $\Gamma_2 = T^{-1}\Gamma_{2,\mathrm{loc}}T$. Then $T(0) = 0$, $T' > 0$, and $S_1^{-1}\Pi_2 S_1(y) = \delta + \Gamma_2(y - \delta)$, proving the formula. The signed root lift is

$$H(t) = t\sqrt{T(t^2)/t^2}, \qquad T(t^2)/t^2 = \int_0^1 T'(st^2)\, ds, \qquad g_2 = H^{-1} g_{2,\mathrm{loc}} H.$$

The quotient is smooth, positive and even. Thus H is odd, increasing, with a smooth bounded inverse on a common root interval; conjugation preserves reversal. Squaring for $t > 0$ gives the transported passage. For $\ell = T'(0)$ its first coefficients are $C_{1,2} = \sqrt{\ell}\, C_{1,2,\mathrm{loc}}$ and $C_{2,2} = \ell C_{2,2,\mathrm{loc}}$, so $C_{2,2} = 3C_{1,2}^2/4$. Nonlinear terms in T first enter the order-$5/2$ remainder. The transported first integral is S_1^{-1} of the old one; its level, first entry and moving first exit follow the same trajectories. The common itinerary therefore gives the chronological first return. Smooth parameter dependence gives the four control functions, and under (2.12) the submersion theorem gives the asserted chart. □

Within an admissible family, finite-time dependence and the implicit function theorem for hitting times give smooth regular transitions with common clock, derivative and domain margins. The opposite strength signs and nonzero closing curvature persist in a neighbourhood; cancellation is imposed only at the reference.

For dependent controls we introduce independent auxiliary parameters for the estimates.

Lemma 2.7 (Independent auxiliary strength and separation)**.** *Let a jointly smooth finite-parameter family have the normalized return*

$$P_\lambda(q) = R_\lambda(\widehat{\delta}(\lambda) + \Gamma_{2,\lambda}(\Gamma_{1,\lambda}(q) - \widehat{\delta}(\lambda))),$$
$$\widehat{m}(\lambda) = C_{1,1}(\lambda) + C_{1,2}(\lambda),$$

with the signed companions of Proposition 2.4. Near a reference with $\widehat{m} = \widehat{\delta} = 0$, there is a jointly smooth root-profile family with independent auxiliary parameters (m, δ), the same coefficient law and the identity $C_{1,1} + \widetilde{C}_{1,2} = m$ for all auxiliary (m, δ), whose return satisfies

$$\widetilde{P}_{\lambda,\widehat{m}(\lambda),\widehat{\delta}(\lambda)} = P_\lambda.$$

The closing map remains R_λ, and all profiles satisfy uniform bounds.

Proof. Put $\iota(t) = -t$, $J_2 = g_{2,\lambda}\iota$ and $\mathcal{N}_2(t, \lambda) = (t - J_2(t, \lambda))/2$. Since $J_2^2 = \mathrm{id}$, $\mathcal{N}_2 J_2 = \iota\mathcal{N}_2$ and $\mathcal{N}_2'(0) = 1$, so $\mathcal{N}_2$ is a jointly smooth increasing local normalizer. For $s = m - \widehat{m}(\lambda)$ set

$$\mathcal{N}_s(t, \lambda) = \mathcal{N}_2(t, \lambda) - \frac{s}{4}t^2, \qquad \widetilde{g}_2 = \mathcal{N}_s^{-1}\iota\mathcal{N}_s\iota.$$

A common shrinking keeps $\mathcal{N}_s' > 0$, and the parameter inverse function theorem gives its joint smooth inverse. Thus $\widetilde{g}_2$ is tangent to the identity, satisfies $\iota\widetilde{g}_2\iota = \widetilde{g}_2^{-1}$, and equals $g_{2,\lambda}$ when $s = 0$. If $\mathcal{N}_2 = t + d_2t^2 + O(t^3)$, the quadratic coefficient of $\mathcal{N}_2^{-1}\iota\mathcal{N}_2\iota$ is $-2d_2$. The change therefore raises the quadratic coefficient of $\widetilde{g}_2$ by $s/2$. Define $\widetilde{\Gamma}_2(x) = x$ for $x \le 0$ and $\widetilde{\Gamma}_2(t^2) = \widetilde{g}_2(t)^2$ for $t > 0$. Its second-passage strength is $\widetilde{C}_{1,2} = C_{1,2} + s$, so $C_{1,1} + \widetilde{C}_{1,2} = \widehat{m} + s = m$. Reversibility gives its coefficient law and smooth root remainder. Replace the second passage in the return by $\widetilde{\Gamma}_2$ and use the independent translation δ. This recovers the original return at $(m, \delta) = (\widehat{m}(\lambda), \widehat{\delta}(\lambda))$. Joint smoothness on compact reductions gives the uniform bounds. □

For the estimates below, a family with dependent controls is first treated in this auxiliary root-profile family and then restricted to $(m, \delta) = (\widehat{m}(\lambda), \widehat{\delta}(\lambda))$. This convention concerns estimates of return maps; it asserts no field realization of the auxiliary profiles.

2.4 Divisions and estimates up to the thresholds

Where both passages are active, the return admits a division in root profiles. To cover the thresholds as well, we also estimate the difference from the aligned return in the coordinate q.

Lemma 2.8 (Division and estimates up to the thresholds)**.** *There are common $r_0 > 0$, $v_0 = 1/4$ and a small parameter neighbourhood in (m, p) such that, for $0 < r \le r_0$, $|v| \le v_0$, $q = r^2$ and $\delta = r^2 v$, both passages are active and the return has the expression*

$$Q - q = mr^3 + \tfrac{3}{4}m^2r^4 + r^5L_0 + \delta rH, \qquad H(0, 0, 0, p) = \tfrac{3}{2}A(p), \tag{2.20}$$

where L_0, H are restrictions of jointly smooth functions of (r, m, v, p) on the signed box $|r| \le r_0$, $|v| \le v_0$, with the same parameter neighbourhood. For arbitrary relative sizes and either sign of q and δ in the common local domain, including the thresholds,

$$|Q - q - mq_+^{3/2}| \le K\{m^2q_+^2 + q_+^{5/2} + |\delta|\sqrt{q_+ + |\delta|} + |\delta|q_+^{3/2}\} \tag{2.21}$$

with a common constant.

Proof. Write $g_\nu(t,\lambda) = tT_\nu(t,\lambda)$ with T_ν jointly smooth and $T_\nu(0,\lambda) = 1$. For signed r and $\delta = r^2 v$, put

$$U = T_1(r,\lambda)^2, \quad W = U - v, \quad s_2 = r\sqrt{W}, \qquad K_0 = v + WT_2(s_2,\lambda)^2.$$

Since $T_1(0,\lambda) = 1$, one common reduction gives $T_1^2 > 3/4$. For $|v| \le v_0 = 1/4$ this implies $W = T_1^2 - v > 1/2$. These expressions are smooth and $Q(r^2) = r^2 K_0$ exactly for $r > 0$. Since $K_0(0,m,v,p) = 1$, joint Hadamard division gives $K_0 - 1 = r\mathcal{Q}$. Aligned cancellation gives $\mathcal{Q}(r,m,0,p) = m + \frac{3}{4}m^2 r + r^2 L_0(r,m,p)$; division in the independent variable v then gives the exact identity

$$\mathcal{Q} = m + \tfrac{3}{4}m^2 r + r^2 L_0 + vH(r,m,v,p), \qquad H(0,0,0,p) = \tfrac{3}{2}A(p). \tag{2.22}$$

Indeed, the first root jet at $r = 0$ is $A_1 + A_2(1-v)^{3/2}$, whose v derivative at cancellation is $-3A_2/2 = 3A(p)/2$. The profile arguments contain $\delta = r^2 v$. Multiplication by r^3 proves (2.20).

Under (2.12), hold (a,b,m,ξ) fixed. For dependent controls, write p for the original physical parameters and hold (p,m) fixed in the auxiliary family of Lemma 2.7, varying only its auxiliary separation δ. After integration, restrict to the observed controls as agreed above. Put $y = \Gamma_1(q) - \delta$. The total derivative is

$$Q_\delta = 1 - \Gamma_{2,q}(y) + \Gamma_{2,q}(y)\Gamma_{1,\delta}(q) + \Gamma_{2,\delta}(y). \tag{2.23}$$

The profile parameter derivatives are $O(x_+^{3/2})$. Integrate (2.23) from zero to δ, with the indicated quantities held fixed. At an intermediate separation t, $y_+ = (\Gamma_1(q,t) - t)_+ \le Kq_+ + |t|$. Joint C^1 regularity and the smooth profiles give, including at active arguments equal to zero,

$$|\Gamma_{\nu,q}(x) - 1| \le K\sqrt{x_+}, \qquad |\Gamma_{\nu,t}(x)| \le Kx_+^{3/2}.$$

Thus $|Q_t| \le K[\sqrt{q_+ + |t|} + q_+^{3/2}]$, and

$$|Q_{m,\delta} - Q_{m,0}| \le K|\delta|[\sqrt{q_+ + |\delta|} + q_+^{3/2}]. \tag{2.24}$$

Together with (2.13) this proves (2.21). □

Remark 2.9 (Checking the hypotheses). With the prescribed contact geometry, transverse coorientation and common first-return itinerary, the remaining conditions involve finitely many jets. The transported strengths are $A_\nu = \frac{4}{3}\mu_\nu(0,0)$; transport from an initial second-port coordinate multiplies its strength by $\sqrt{S_1'(\delta)}$, as in Proposition 2.6. At the reference, check $A_1 = -A_2 \neq 0$, $R(0) = 0$, $R'(0) = 1$ and $c_0 = R''(0)/2 \neq 0$. These conditions give admissibility and the upper bounds of Theorem 1.2 and part 1 of Theorem 1.3. Attainment additionally uses (2.12); the saturation and fourfold-cycle conclusions use the sign sector specified in Theorems 1.3 and 1.4.

The curvature of this composition gives the count in the next section, and the value estimates locate the six-cycle configurations.

3 Sharp cyclicity

We count fixed points by following the zeros of P'', which we call the curvature of the return. Where both passages are active and the state is large compared with the separation (the outer region), these zeros lie on a separation graph with at most one positive critical point; estimates near the thresholds (the threshold layer) fix the sign of P'' there. Together they give the

upper bounds on the whole interval I. We then construct configurations attaining the bounds under (2.12).

An active three-halves term has curvature proportional to $q^{-1/2}$. In the outer region, where both passages are active, $q = r^2$ and $J = rP''$ remove this singular factor, while $v = \delta/q$ measures separation relative to the state. With the aligned multiplier $k(p) = 1 + b > 0$, their first-order balance is

$$J = 2c_0 r + \frac{3k}{4}m - \frac{3kA}{8}v + O\big((r + |m| + |v|)^2\big).$$

The strength $A \neq 0$ makes the curvature equation transverse in v. Exact division then gives a separation graph Δ with $\Delta_r = r\mathcal{W}$; the closing curvature $c_0 \neq 0$ makes $\mathcal{W}$ strictly monotone. These two reductions give at most one positive critical point.

3.1 The exact curvature-zero graph

Let g_ν be the signed-root companion in Proposition 2.4. Its dependence on (p, m, δ) is suppressed in the notation. Define smooth functions

$$T_\nu(u) = g_\nu(u)/u, \quad \ell_\nu(u) = T_\nu(u)^2 + \tfrac{u}{2}(T_\nu^2)'(u), \quad H_\nu(u) = \tfrac{1}{2}\ell_\nu'(u). \tag{3.1}$$

The quotient is filled in smoothly at $u = 0$, with $T_\nu(0) = 1$. For $u > 0$,

$$\Gamma_\nu'(u^2) = \ell_\nu(u), \qquad \Gamma_\nu''(u^2) = u^{-1}H_\nu(u), \qquad \ell_\nu(0) = 1, \quad H_\nu(0) = \tfrac{3}{4}A_\nu. \tag{3.2}$$

These are exact identities obtained by differentiating $\Gamma_\nu(u^2) = g_\nu(u)^2$.

For $r > 0$, put

$$q = r^2, \quad \delta = r^2 v, \quad J(r, m, v, p) = rP''_{m,r^2v}(r^2, p). \tag{3.3}$$

Choose a fixed small $0 < \eta \le v_0 = 1/4$, within the box of Lemma 2.8. All profiles in the following calculation are evaluated at $\delta = r^2 v$. Write

$$U = T_1(r)^2, \qquad W = U - v, \qquad s_2 = r\sqrt{W}.$$

After a fixed common shrinking, $W > 1/2$ for $|v| \le \eta$. The two excursions are then both active and the exact downstream penetration is $y = r^2 W > 0$. Chain differentiation gives

$$Q = r^2\{v + WT_2(s_2)^2\}, \qquad Q' = \ell_2(s_2)\ell_1(r), \tag{3.4}$$

$$Q'' = r^{-1}\{\ell_2(s_2)H_1(r) + W^{-1/2}H_2(s_2)\ell_1(r)^2\}. \tag{3.5}$$

Consequently

$$\begin{aligned} J = {} & rR''(Q)[\ell_2(s_2)\ell_1(r)]^2 \\ & + R'(Q)\{\ell_2(s_2)H_1(r) + W^{-1/2}H_2(s_2)\ell_1(r)^2\}. \end{aligned} \tag{3.6}$$

Every expression on the right is a root profile, jointly smooth through $r = 0$.

Lemma 3.1 (Root curvature and its zero graph)**.** *The function J has a joint smooth extension on a common box in (r, m, v, p). At its root boundary,*

$$J(0, m, v, p) = \tfrac{3}{4}R'_{p,m,0}(0)\{A_1(p, m, 0) + A_2(p, m, 0)(1 - v)^{-1/2}\}. \tag{3.7}$$

In particular,

$$J_v(0, 0, 0, p) = -\tfrac{3}{8}R'_{p,0,0}(0)A(p) \neq 0. \tag{3.8}$$

There is exactly one smooth graph $v = V(r, m, p)$, with $|V| < \eta$, on which $J = 0$ in this box.

Proof. Formula (3.6) proves smoothness, and (3.2) at $r = s_2 = 0$ gives (3.7). Differentiation in v gives (3.8). First choose η so that J_v has the strict sign $-A$ and a uniform nonzero margin on the root boundary. The signs of $J(0, 0, -\eta, p)$ and $J(0, 0, \eta, p)$ are opposite. Compactness in the remaining parameters permits one further shrinking of r, m on which these properties persist throughout the box. The implicit function theorem, existence by the endpoint signs, and strict monotonicity in v give the asserted unique graph. □

The boundary value of the graph is exact:

$$V(0, m, p) = 1 - \left(\frac{-A_2(p, m, 0)}{A_1(p, m, 0)}\right)^2, \qquad V_m(0, 0, p) = \frac{2}{A(p)}. \tag{3.9}$$

Indeed the ratio equals $1 - m/A_1(p, m, 0)$; both individual strengths retain their dependence on m.

To bound the positive critical points of the separation graph $r \mapsto r^2 V(r, m, p)$ on a fixed interval, divide first along the cancelling slice and then in the independent strength variable. Finite cancellation gives

$$Q_{0,0}(q, p) = q + q_+^{5/2} L(\sqrt{q_+}, p). \tag{3.10}$$

For positive $q = r^2$ it follows, by differentiation in q, that

$$P''_{0,0}(r^2, p) = 2c_0(p) + rN(r, p), \qquad c_0(p) = \tfrac{1}{2} R''_{p,0,0}(0), \tag{3.11}$$

where N is jointly smooth. The offset and linear jets are allowed to vary in p throughout this identity.

Let $V_0(r, p) = V(r, 0, p)$. Since $J(r, 0, V_0, p) = 0$, the mean-value identity in the independent variable v yields

$$0 = r\{2c_0 + rN\} + V_0 \int_0^1 J_v(r, 0, tV_0, p)\, dt.$$

The integral has a nonzero common margin. Therefore $V_0 = rB_\Delta(r, p)$ with B_Δ smooth and

$$B_\Delta(0, p) = \frac{16c_0(p)}{3R'_{p,0,0}(0)A(p)}. \tag{3.12}$$

Hadamard division in m now gives the exact identity

$$V(r, m, p) = mH_\Delta(r, m, p) + rB_\Delta(r, p), \qquad H_\Delta(0, 0, p) = \frac{2}{A(p)}. \tag{3.13}$$

Define the separation graph

$$\Delta(r, m, p) = r^2 V(r, m, p). \tag{3.14}$$

Its derivative factors exactly:

$$\Delta_r = r\mathcal{W}, \qquad \mathcal{W} = 2V + rV_r. \tag{3.15}$$

The boundary coefficients in (3.13) give

$$\mathcal{W}_m(0, 0, p) = \frac{4}{A(p)}, \qquad \mathcal{W}_r(0, 0, p) = 3B_\Delta(0, p) = \frac{16c_0(p)}{R'_{p,0,0}(0)A(p)} \neq 0. \tag{3.16}$$

Joint smoothness and compactness in p give a common smaller box on which $\mathcal{W}_r$ has a uniform nonzero margin with sign c_0/A.

Proposition 3.2 (At most one critical point of the separation graph)**.** *For fixed (m, p), the graph $r \mapsto \Delta(r, m, p)$ has at most one positive critical point on a common interval. Any such point is nondegenerate. In the outer region $|\delta|/r^2 \le \eta$,*

$$P''_{m,\delta}(r^2, p) = r^{-3}\mathcal{U}(r, m, \delta/r^2, p)[\delta - \Delta(r, m, p)], \tag{3.17}$$

where $\mathcal{U}$ is jointly smooth and nonzero, of sign $-A$. For fixed parameters the zeros of P'' in the outer region have total multiplicity at most two.

Proof. For fixed (m, p), $\mathcal{W}$ is strictly monotone and has at most one positive zero. At that zero, $\Delta_{rr} = r\mathcal{W}_r \neq 0$. Hence Δ has at most two monotone branches.

For the factorization, define

$$\mathcal{U}(r, m, v, p) = \int_0^1 J_v(r, m, V + t(v - V), p)\, dt.$$

The derivative bound from Lemma 3.1 gives its sign and nonvanishing. The exact mean-value identity is $J = (v - V)\mathcal{U}$. Substituting $v = \delta/r^2$ in (3.3) gives (3.17). A horizontal line meets this graph in at most two transverse points, or at its sole positive critical point. Transverse intersections are simple; an intersection at the nondegenerate critical point has order two. Changing from $r > 0$ to $q = r^2$ is a smooth local diffeomorphism, and the prefactor in (3.17) never vanishes. This proves the multiplicity assertion as well as the distinct count. □

We now treat the complement of the outer region, where an active input may be arbitrarily close to its threshold.

3.2 The count on the whole interval

Use the thresholds 0 and $a_\delta = \Gamma_1^{-1}(\delta)$ from (2.15), with the active intervals determined there. All constants are uniform on the fixed parameter reduction.

Where both passages are active in the threshold layer, compare the difference of the singular curvatures with their sum. Separation keeps the active inputs at a fixed relative distance; the difference then dominates both the relative error and the bounded closing curvature.

Put $\vartheta = |\delta|$, $\sigma = \operatorname{sgn}\delta$, and let $L \ge 2$ be fixed. The strengths satisfy, uniformly,

$$A_1(p, m, \delta) = A(p) + O(|m| + \vartheta), \qquad A_2(p, m, \delta) = -A(p) + O(|m| + \vartheta).$$

The smooth root profiles in (2.5) yield

$$\Gamma'_\nu(x) = 1 + \tfrac{3}{2}A_\nu\sqrt{x} + O(x), \tag{3.18}$$

$$\Gamma''_\nu(x) = \tfrac{3}{4}A_\nu x^{-1/2} + O(1), \qquad x > 0. \tag{3.19}$$

Differentiating the smooth root profiles gives these remainders and their required derivative bounds. Each Γ'_ν is uniformly positive and bounded, with active curvature of the sign of its strength.

For $0 < q \le L\vartheta$,

$$|\Gamma_1(q) - q| \le Kq^{3/2} \le KL^{3/2}\vartheta^{3/2}. \tag{3.20}$$

After reducing ϑ, this is at most $\vartheta/2$. The inverse derivative estimates give $\vartheta/2 < a_\vartheta < 2\vartheta$. Every positive penetration in this layer is at most $(L + 2)\vartheta$.

Lemma 3.3 (Curvature signs in the threshold layer)**.** *For every fixed $L \ge 2$, one may choose common sufficiently small bounds on $|m|$ and $\vartheta = |\delta|$ such that, for $0 < |\delta|$, the signs of P'' are*

$$\operatorname{sgn} P'' = \begin{cases} \operatorname{sgn} c, & q < \min(0, a_\delta), \\ \operatorname{sgn}(\delta A), & \min(0, a_\delta) < q < \max(0, a_\delta), \\ \operatorname{sgn}(-\delta A), & q > \max(0, a_\delta), \quad q \le L|\delta|. \end{cases} \tag{3.21}$$

Both passages are active on the last interval. There is no zero of P'' in any of these open regions. In the active regions $|Q''| \geq \kappa_L \vartheta^{-1/2}$ for a uniform $\kappa_L > 0$. The estimates are uniform up to the thresholds.

Proof. Below both thresholds both passages are inactive, so $Q = q$ and $P'' = R''$, with sign c.

If $\delta = \vartheta > 0$, the one-active interval is $0 < q < a_\vartheta$, where $Q = \Gamma_1(q)$. Its curvature has sign A, and (3.19) gives $|Q''| \geq \kappa_L (2\vartheta)^{-1/2}$ after shrinking. If $\delta = -\vartheta < 0$, that interval is $-\vartheta < q < 0$, where $Q = -\vartheta + \Gamma_2(q + \vartheta)$. Its curvature has sign $-A$ and the same uniform lower bound.

Where both passages are active, put

$$y = \Gamma_1(q) - \delta > 0.$$

The exact chain rule is

$$Q'' = \Gamma_2'(y)\Gamma_1''(q) + \Gamma_2''(y)\Gamma_1'(q)^2.$$

Equations (3.18)–(3.20) therefore give

$$Q'' = \tfrac{3}{4} A(q^{-1/2} - y^{-1/2}) + E, \qquad |E| \leq C_L(\sqrt{\vartheta} + \vartheta + |m|)(q^{-1/2} + y^{-1/2}). \tag{3.22}$$

Each error is measured relative to the sum of the two singular weights, so the formula remains uniform as either positive argument tends to zero.

By (3.20), $\sigma(q - y) \geq \vartheta/2$ and $q, y \leq (L + 2)\vartheta$. Hence

$$-\sigma \frac{q^{-1/2} - y^{-1/2}}{q^{-1/2} + y^{-1/2}} = \frac{|q - y|}{(\sqrt{q} + \sqrt{y})^2} \geq \frac{1}{8(L + 2)} =: d_L > 0.$$

First reduce $|m|$, and then ϑ, so the error in (3.22) is less than half the principal bound. Thus $\operatorname{sgn} Q'' = \operatorname{sgn}(-\sigma A)$ and $|Q''| \geq \kappa_L \vartheta^{-1/2}$, throughout the open two-active layer.

Finally the exact closing-leg chain rule is

$$P'' = R''(Q)(Q')^2 + R'(Q)Q''.$$

The first term is uniformly bounded, whereas R' has a positive margin and the preceding active curvature bound diverges as $\vartheta \downarrow 0$. A further fixed reduction proves (3.21). □

Proposition 3.4 (Upper bound on the whole interval)**.** *On a sufficiently small interval $I = (-q_0, q_0)$, common to the family, every return map P_λ, $\lambda \in U$, has at most six fixed points.*

Proof. First choose the outer box and its fixed η in Lemma 3.1. Set $L = 1/\eta \geq 2$, shrinking η if needed. Apply Lemma 3.3 with this one L: reduce the common strength bound and then the separation bound. All further reductions are made inside the first-return domain of Definition 2.1. Choose q_0, δ_0 so both thresholds remain strictly inside $I = (-q_0, q_0)$, for example using $\delta_0 < q_0/4$ and the uniform inverse derivative bound.

For $\delta \neq 0$, P'' has no zeros in the three regions of (3.21). Any remaining zero of P'' has $q > L|\delta| > 0$, so it lies in the outer region $|\delta|/q < \eta$. Proposition 3.2 exhausts this region and gives at most two zeros of P''. At each threshold P'' has a singularity with nonzero coefficient, so a threshold is not a zero of P''.

The derivative P' is continuous at the thresholds. It is strictly monotone on at most five consecutive pieces: the two thresholds and at most two zeros of P'' are the only possible divisions. If the signs of P'' on the two sides of a threshold agree, monotonicity extends through the threshold; if a zero of P'' does not change sign, it adds no new piece. There is no interval on which P'' vanishes. Thus $P' - 1$ has at most five zeros. Rolle's theorem for the C^1 displacement $D = P - q$, including at the thresholds, gives at most six zeros.

The case $\delta = 0$ follows from the next corollary. □

Corollary 3.5 (The four-point slices)**.** *On the same interval the bound on the number of fixed points is four when $\delta = 0$ or $m = 0$, and two when $m = \delta = 0$.*

Proof. For $\delta = 0$, a positive zero of P'' has $V(r, m, p) = 0$. Equations (3.12)–(3.13) give

$$V_r(0, 0, p) = B_\Delta(0, p) = \frac{16c_0(p)}{3k(p)A(p)} \neq 0.$$

Compactness in p gives a common nonzero margin for V_r on a smaller box. Thus, for fixed m, p, there is at most one positive zero of V, and it is simple. Formula (3.17) and the local diffeomorphism $q = r^2$, $r > 0$, preserve its order. Together with at most one sign change at the aligned threshold, this gives at most three monotonicity pieces for P' and four fixed points.

For $m = 0, \delta \neq 0$, the separation graph is $\Delta = r^3 B_\Delta(r, p)$, with sign c_0/A and nonzero derivative. It has at most one curvature intersection, necessarily simple, and one is possible only if $c_0 \delta A > 0$. When this inequality holds, the first two signs in (3.21) agree, leaving one threshold change and at most one outer change. If $c_0 \delta A < 0$, there is no outer intersection and at most two threshold changes. Again P' has at most three monotonicity pieces.

At $m = \delta = 0$, equation (3.11) and the inactive closing map give the same strict sign c_0 throughout the interval. The displacement is strictly convex or strictly concave, including across $q = 0$, and has at most two zeros. □

Corollary 3.6 (Open six-cycle regions)**.** *In every admissible family, the parameters $\lambda \in U$ for which P_λ has six simple fixed points in I form an open set.*

Proof. Choose six disjoint compact intervals isolating the simple zeros of D. Joint C^1 dependence, including at the thresholds, preserves the opposite endpoint signs and the nonzero derivative signs for nearby parameters, so each interval keeps one simple zero. Proposition 3.4 excludes further zeros in I. □

Propositions 3.9 and 3.10 below show that all four bounds are attained under (2.12).

3.3 The cubic critical value

When the separation graph has a positive critical point, its height determines the two-intersection window. We compute this height and its cubic separation scale for the saturation argument.

Proposition 3.7 (The critical value of the separation graph)**.** *After a common further reduction, the signed critical point of the separation graph is*

$$r_c(m, p) = -\frac{k(p)m}{4c_0(p)} + O(m^2), \qquad r_c(0, p) = 0.$$

It is smooth through zero. A positive critical point exists exactly when $mc_0(p) < 0$, and then it equals $r_c(m, p)$. Its critical value has a smooth division

$$\delta_c(m, p) = \Delta(r_c(m, p), m, p) = m^3 \kappa_c(m, p), \qquad \kappa_c(0, p) = \frac{k(p)^2}{24A(p)c_0(p)^2}. \tag{3.23}$$

The remainder and all reductions are uniform in p. On a fixed interval $0 < r < \sqrt{q_0}$, reduced as below, P'' has exactly two simple zeros in the outer region if and only if

$$mc_0(p) < 0, \qquad 0 < \delta/\delta_c(m, p) < 1. \tag{3.24}$$

At $\delta = \delta_c$, $mc_0 < 0$, it instead has one double zero in the outer region.

Proof. Apply the implicit function theorem to $\mathcal{W} = 0$. The boundary derivatives (3.16) and the uniform margin give a unique smooth signed solution $r_c = m\gamma$, with $\gamma(0,p) = -k/(4c_0)$. Its positive part exists precisely for $mc_0 < 0$. Substitution into $\Delta = r^2(mH_\Delta + rB_\Delta)$ gives the smooth quotient

$$\kappa_c = \gamma^2\{H_\Delta(m\gamma, m, p) + \gamma B_\Delta(m\gamma, p)\}.$$

The boundary coefficients yield $\gamma B_\Delta = -4/(3A)$, hence (3.23). Thus κ_c has the sign of A; when a positive critical point exists, the critical value has sign $-c_0/A$.

First fix $r_0 = \sqrt{q_0}$ inside the common root box. Then reduce $|m|$, uniformly in p, so $0 < r_c < r_0/2$ when $mc_0 < 0$, and

$$|mH_\Delta(r_0, m, p)| < \tfrac{1}{2} r_0 |B_\Delta(r_0, p)|.$$

Thus $\Delta(r_0, m, p)$ has sign c_0/A, opposite to the nonzero critical value. Also require $|\delta_c|$ to lie within the common separation bound. The unique positive critical point and $\Delta(0) = 0$ give exactly two transverse intersections with the line δ precisely for (3.24). At each intersection $|\delta|/r^2 = |V(r, m, p)| < \eta$, so both lie in the outer region. Conversely two distinct outer intersections force the unique positive critical point between them and place δ strictly between zero and its critical value. The factorization (3.17) transfers their orders to P''. At the critical value, the nondegenerate critical point gives $D'' = D''' = 0$ and $D'''' \neq 0$. □

On the curve $\delta = \delta_c$ the conditions $D = D' = 0$ are not imposed, so this curve is not a fold curve of periodic orbits.

Figure 2 shows a separation graph with one positive critical point.

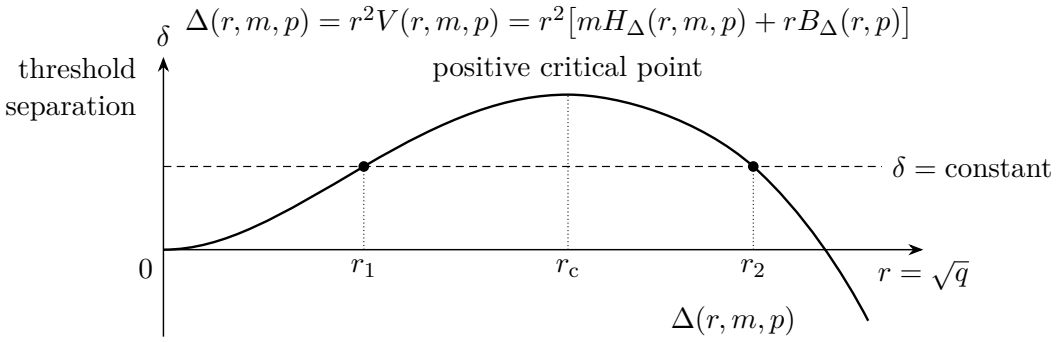


Figure 2: The separation graph $\delta = \Delta(r, m, p)$ for fixed (m, p) with $c_0 < 0$, $A > 0$ and $m > 0$ (schematic). The graph has at most one positive critical point, which is nondegenerate (Proposition 3.2). Its intersections with a horizontal line are the zeros of P'' in the outer region $q > L|\delta|$, by (3.17); for other parameters the graph can be monotone. The resulting bound on fixed points is Proposition 3.4.

3.4 Attainment of the bounds

To attain six fixed points we choose a quartic with four simple positive roots; its coefficients determine a quadratic with two simple negative roots. We continue the two groups of roots in disjoint state intervals and then treat the slices $m = 0$ and $\delta = 0$.

Assume (2.12) and use the standard unfolding $\lambda = \Lambda(a, b, m, \delta)$ of Section 1, with $\lambda_0 = \Lambda(0, 0, 0, 0)$. Write

$$A_0 = A_1(\lambda_0), \quad c_0 = \tfrac{1}{2} R''_{\lambda_0}(0), \quad B = |A_0|, \quad C = |c_0|,$$

$$s = -\operatorname{sgn} c_0, \qquad \tau = \operatorname{sgn}(sA_0).$$

We first allow an arbitrary rescaled strength sum, including zero:

$$\lambda_\varepsilon = \Lambda(\bar{a}\varepsilon^4, \bar{b}\varepsilon^2, \bar{m}\varepsilon, \bar{d}\varepsilon^3), \qquad \theta = (\bar{a}, \bar{b}, \bar{m}, \bar{d}), \qquad \varepsilon > 0. \tag{3.25}$$

Lemma 3.8 (Rescaled displacement)**.** *Let $\Omega_+ \Subset (0,\infty)$, $\Omega_- \Subset (-\infty, 0)$, and let θ range over a fixed compact set. For sufficiently small $\varepsilon > 0$, the rescaled displacements satisfy, for $z \in \Omega_+$ and $x \in \Omega_-$,*

$$\mathcal{F}_\varepsilon(z) := \varepsilon^{-4} D_{\lambda_\varepsilon}(\varepsilon^2 z^2) = F_\theta(z) + \varepsilon E_+(\varepsilon, z, \theta), \tag{3.26}$$
$$F_\theta(z) = \bar{a} + \bar{b}z^2 + \bar{m}z^3 + c_0 z^4 + \tfrac{3}{2}A_0 \bar{d} z,$$
$$\mathcal{G}_\varepsilon(x) := \varepsilon^{-4} D_{\lambda_\varepsilon}(\varepsilon^2 x) = G_\theta(x) + \varepsilon E_-(\varepsilon, x, \theta), \tag{3.27}$$
$$G_\theta(x) = \bar{a} + \bar{b}x + c_0 x^2.$$

The remainders extend jointly smoothly to signed ε near zero. For every prescribed finite integer n, uniformly on these compacts,

$$\max_{j+|\gamma|\le n} \left\| \partial_z^j \partial_\theta^\gamma (\mathcal{F}_\varepsilon - F_\theta) \right\|_\infty \le C_n \varepsilon, \tag{3.28}$$

and similarly with $x, \mathcal{G}_\varepsilon$. The compact parameter set may contain points with $\bar{m} = 0$.

Proof. Use the exact identities of Lemma 2.8. Substitute $r = \varepsilon z$, $v = \bar{d}\varepsilon/z^2$ into (2.22); the profiles are evaluated at $\delta = r^2 v = \bar{d}\varepsilon^3$. Then

$$Q - q = \varepsilon^4 \Xi, \qquad \Xi = \bar{m}z^3 + \bar{d}zH + \varepsilon z^5 L_0 + \tfrac{3}{4}\bar{m}^2\varepsilon^2 z^4.$$

With $Y = z^2 + \varepsilon^2 \Xi$, the closing expansion gives

$$\mathcal{F}_\varepsilon = \bar{a} + \bar{b}Y + \Xi + c(\lambda_\varepsilon)Y^2 + \varepsilon^2 Y^3 S(\varepsilon^2 Y, \lambda_\varepsilon). \tag{3.29}$$

Since z stays away from zero, all arguments remain in the common root box, whose positive square-root factor stays bounded away from zero. At $\varepsilon = 0$, $H = 3A_0/2$ and $c = c_0$, giving F_θ. Taylor integral division in ε proves the joint remainder and every finite mixed estimate in (3.28), without dividing by $\bar{m}$. State derivatives are taken at fixed (ε, θ), hence at fixed λ_ε; the variable $v = \bar{d}\varepsilon/z^2$ keeps the separation $r^2 v = \bar{d}\varepsilon^3$ fixed.

For negative x in a fixed compact set, $q = \varepsilon^2 x$ is below both thresholds for small positive ε, since $\delta = O(\varepsilon^3)$ and $|a_\delta| \le 2|\delta|$. Thus $Q = q$ exactly, and (2.11) gives

$$\mathcal{G}_\varepsilon = \bar{a} + \bar{b}x + c(\lambda_\varepsilon)x^2 + \varepsilon^2 x^3 S(\varepsilon^2 x, \lambda_\varepsilon).$$

This proves the negative formula and its joint smooth extension. □

For the six-point construction specialize to $\bar{a} = s\alpha_4$, $\bar{b} = s\beta_4$, $\bar{m} = s$ and $\bar{d} = \tau d_4$, giving

$$m = s\varepsilon, \qquad \delta = \tau d_4 \varepsilon^3, \qquad b = s\beta_4\varepsilon^2, \qquad a = s\alpha_4\varepsilon^4, \qquad \varepsilon > 0. \tag{3.30}$$

Here α_4, β_4, d_4 are rescaled parameters. Multiplying the rescaled displacements by s, and using $s\tau A_0 = B$, $sc_0 = -C$, gives $\mathcal{F}_{4,\varepsilon} := s\mathcal{F}_\varepsilon = F_0 + O(\varepsilon)$ and $\mathcal{G}_{4,\varepsilon} := s\mathcal{G}_\varepsilon = G_0 + O(\varepsilon)$, where

$$F_0(z) = \alpha_4 + \beta_4 z^2 + z^3 - Cz^4 + \tfrac{3}{2}Bd_4 z,$$
$$G_0(x) = \alpha_4 + \beta_4 x - Cx^2.$$

These expansions have all the mixed estimates of Lemma 3.8.

Take distinct positive numbers r_i, $1 \le i \le 4$, with $\sum r_i = 1/C$, and let e_j be their elementary symmetric functions. Choosing

$$\alpha_4 = -Ce_4, \qquad \beta_4 = -Ce_2, \qquad d_4 = \frac{2C}{3B}e_3 \tag{3.31}$$

gives simultaneously

$$F_0(z) = -C\prod_{i=1}^{4}(z - r_i), \qquad G_0(x) = -C(x^2 + e_2x + e_4).$$

The six pair products have product e_4^3; AM–GM yields $e_2 \ge 6\sqrt{e_4}$. Thus $e_2^2 - 4e_4 > 0$, and the quadratic has two distinct negative roots. Hence the choice (3.31) gives four simple positive roots of F_0 and two simple negative roots of G_0.

Proposition 3.9 (Six fixed points)**.** *For every admissible family satisfying (2.12), the standard unfolding has six simple fixed points of order ε^2 for all parameters (3.30) with rescaled parameters in an open set and all sufficiently small $\varepsilon > 0$. Their stability alternates: three are attracting and three repelling.*

Proof. Choose any distinct positive r_i with sum $1/C$ and the parameters (3.31). The four roots of F_0 and the two negative roots of G_0 have strict derivative margins. Lemma 3.8 continues them on six disjoint isolating intervals, uniformly over a small open set of rescaled parameters. The same joint estimates allow $\bar{m}$ to vary near s, so this set may be taken open in all four rescaled parameters. Fixed rescaled parameters give a smooth parameter path tending to the reference, and Proposition 3.4 excludes additional fixed points. At these parameters, for negative states $D_q = s\varepsilon^2 G_0'(x) + O(\varepsilon^3)$, and for positive states $D_q = s\varepsilon^2 F_0'(z)/(2z) + O(\varepsilon^3)$. In state order these signs are $s(+,-,+,-,+,-)$. Since the return is increasing and close to the identity, they give three attracting and three repelling hyperbolic cycles; these signs persist on the open set of rescaled parameters. □

Proposition 3.10 (Attainment on the slices)**.** *For every admissible family satisfying (2.12), the standard unfolding has four simple fixed points on $\delta = 0$, four on $m = 0$, and two on $m = \delta = 0$. Each configuration occurs in I for every sufficiently small $\varepsilon > 0$ along a smooth parameter path tending to the reference, and persists on an open set of rescaled parameters within the corresponding slice.*

Proof. Use the weights (3.25). The following table specifies the paths and the limiting simple roots; positive roots use the coordinate z, and negative roots the coordinate x:

slice	$(\bar{a}, \bar{b}, \bar{m}, \bar{d})/(sC)$	$z > 0$	$x < 0$
$\delta = 0$	$(36/11, -85/11, 60/11, 0)$	$1, 2, 3$	x_δ
$m = 0$	$(36, 25, 0, -40/A_0)$	$1, 2, 3$	x_m
$m = \delta = 0$	$(1, 0, 0, 0)$	1	-1

(3.32)

Here $x_\delta = (-85 - \sqrt{8809})/22$ and $x_m = (25 - \sqrt{769})/2$. For the first two rows, direct substitution in Lemma 3.8 gives respectively

$$F_\theta/(sC) = -(z-1)(z-2)(z-3)(z+6/11), \quad G_\theta/(sC) = 36/11 - 85x/11 - x^2,$$
$$F_\theta/(sC) = -(z-1)(z-2)(z-3)(z+6), \qquad G_\theta/(sC) = 36 + 25x - x^2.$$

Only the positive roots of F_θ correspond to fixed points; the negative ones come from G_θ. The estimates of Lemma 3.8 include $\bar{m} = 0$.

For the third row, (2.17) and the root-profile remainder estimates give, uniformly on compact x-intervals including zero,

$$\frac{D_{\lambda_\varepsilon}(\varepsilon^2 x)}{sC\varepsilon^4} = 1 - x^2 + O_{C^1}(\varepsilon).$$

Continue the listed simple roots by Lemma 3.8 and the isolating argument in the proof of Proposition 3.9 on small relatively open sets of rescaled parameters with compact closures: $\bar{d} = 0$ in the first row, $\bar{m} = 0$ in the second, and $\bar{m} = \bar{d} = 0$ with $(\bar{a}, \bar{b})$ near $(sC, 0)$ in the third. The rescaled thresholds are 0 and a_δ/ε^2, with $|a_\delta|/\varepsilon^2 \le 2|\bar{d}|\varepsilon$. One reduction of ε places the positive intervals where both passages are active, the negative ones below both thresholds, the parameters in the domain of Λ, and the rescaled intervals in I. Corollary 3.5 excludes other roots in I, including at the thresholds, and Λ gives smooth parameter paths and relative persistence. □

The positive quartet in the six-cycle construction is the starting point for Section 4.

4 The saturation domain at the principal scale

The saturation domain has a limiting geometry described by a quartic with four nonnegative roots. For a return with six fixed points, we first derive the weights of the parameters and of the fixed points from the five alternating extrema of its displacement. Estimates near the thresholds then control critical points that approach the thresholds, and their limiting values determine the four-root region. The necessary estimates hold in every admissible family. We prove the weights introduced in Section 1, beginning with a tail estimate that converts weighted parameter bounds into bounds on the fixed points.

Lemma 4.1 (A tail bound from weighted parameters)**.** *For every fixed $K_0 > 0$, there are common constants $M, m_0 > 0$ such that, whenever $0 < |m| < m_0$ and*

$$|a| \le K_0 m^4, \qquad |b| \le K_0 m^2, \qquad |\delta| \le K_0 |m|^3,$$

every zero of D in I satisfies $|q| < Mm^2$. On the two complementary tails, D has the sign of c.

Proof. Let $0 < q_{\mathrm{in}} < q_0$ be an inner radius, to be fixed below. For $Mm^2 \le q \le q_{\mathrm{in}}$, reduce m_0 so that $|\delta|/q \le K_0|m|/M \le 1$. Equation (2.21) gives

$$\frac{|Q - q|}{q^2} \le K\{M^{-1/2} + M^{-3/2} + m^2 + \sqrt{q_{\mathrm{in}}} + m^2 M^{-1/2}\}.$$

In particular, the strength-sum and separation terms use respectively $|m|/\sqrt{q} \le M^{-1/2}$ and $|\delta|\sqrt{q + |\delta|}/q^2 \le KM^{-3/2}$. For small q_{in}, this also gives $|Q - q|/q < 1/2$. The closing expansion yields

$$D - cq^2 = a + bq + (1 + b)(Q - q) + c\{2q(Q - q) + (Q - q)^2\} + Q^3 S(Q, \lambda).$$

Consequently

$$\frac{|D(q) - c(\lambda)q^2|}{q^2} \le K\{M^{-2} + M^{-1} + M^{-1/2} + M^{-3/2} + \sqrt{q_{\mathrm{in}}} + |m|\}. \tag{4.1}$$

The first two terms bound a/q^2 and b/q; the remaining closing terms follow from $|Q| \le 2q$ and the bound for $Q - q$. For $-q_{\mathrm{in}} \le q \le -Mm^2$, both passage inputs are negative once m_0 is

small: $q < \min(0, \delta)$. Thus $Q = q$ exactly, and the same estimate follows directly from the closing expansion. Choose M large, then q_{in} small and m_0 small so that the right-hand side is less than half the common lower bound for $|c|$; reduce m_0 further so that $Mm_0^2 < q_{\text{in}}$.

The rest of the original common interval $I = (-q_0, q_0)$ is covered as follows. At the reference, (2.17) gives $D(0) = D'(0) = 0$ and strict curvature of sign c_0 throughout the reduced interval. Hence D has that sign at every nonzero reference state. The closed outer state set $q_{\text{in}} \le |q| \le q_0$ is compact and has a nonzero value margin; joint continuity in the parameters preserves it after one common parameter reduction. The common domains are chosen to contain $\overline{I}$, so this argument covers the whole interval. All choices are made before any parameter path is chosen. □

Lemma 4.2 (Returns with six fixed points)**.** *If D has six zeros in I, then*

$$mc_0(p) < 0, \qquad \delta A(p)c_0(p) < 0, \qquad 0 < \delta/\delta_{\text{c}}(m, p) < 1.$$

In particular $0 < |\delta| < |\delta_{\text{c}}| \le K|m|^3$ uniformly. Put $s = -\operatorname{sgn} c_0$, $E = sD$, and denote the thresholds by $\theta_1 < \theta_2$. Then P'' has two simple zeros $\zeta_1 < \zeta_2$ in the outer region, and

q	$(-q_0, \theta_1)$	(θ_1, θ_2)	(θ_2, ζ_1)	(ζ_1, ζ_2)	(ζ_2, q_0)
$\operatorname{sgn} E''$	$-$	$+$	$-$	$+$	$-$

(4.2)

The continuous E' has exactly one zero in each open interval, none at its shared junctions, and

$$E'(\theta_1) < 0, \quad E'(\theta_2) > 0, \quad E'(\zeta_1) < 0, \quad E'(\zeta_2) > 0. \tag{4.3}$$

All six zeros are simple C^1 zeros, including any at a threshold. The extremal values have signs $+, -, +, -, +$; the first is an inactive positive maximum and the second a negative minimum between the thresholds. On $\delta = \delta_{\text{c}}$, $mc_0 < 0$, there are at most four fixed points.

Proof. The four-point slices exclude $m = 0$ and $\delta = 0$. The signs (3.21) give the number $s_E = 1 + \mathbf{1}_{\{\delta A c_0 < 0\}}$ of curvature sign changes at the thresholds. Let $s_O \le 2$ be the number of sign changes at zeros of P'' in the outer region. Joining same-sign pieces through isolated zeros and thresholds gives at most $s_E + s_O + 1 \le 5$ strict monotonicity blocks for the continuous D'. Each block, with its finite endpoints, contains at most one zero. Six zeros of D give five distinct derivative zeros in their open gaps by C^1 Rolle, so $s_E = s_O = 2$. Proposition 3.7 gives the stated window and cubic bound. The signs (3.21) and the outer factorization (3.17) give (4.2).

Retain every shared finite junction in both adjacent blocks of E'. A derivative zero there would occupy both blocks, leaving room for at most four distinct derivative zeros. Hence all five lie in the five respective interiors, and strict monotonicity gives (4.3). A stationary fixed point would supply a sixth derivative zero distinct from the five open-gap zeros. Thus none occurs, including at a threshold. The extrema interlace the six roots, giving all five strict value signs. At $\delta = \delta_{\text{c}}$ the double zero of P'' does not change sign, leaving at most three blocks and four distinct fixed points. Truncating at the ends of the common interval removes blocks but cannot add them. □

Proposition 4.3 (Weighted confinement)**.** *For every admissible family there are a further reduction of I and U and a constant $K > 0$ with the following property. If P_λ has six fixed points $q_1, \dots, q_6$ in I, then*

$$0 < |\delta| \le K|m|^3, \qquad |b| \le K|m|^2, \qquad |a| \le K|m|^4, \qquad \max_j |q_j| \le K|m|^2. \tag{4.4}$$

Proof. We use the auxiliary-control convention following Lemma 2.7 also when m and δ are dependent. The jets a, b are those of (2.10).

Lemma 4.2 already gives the first bound in (4.4). At a curvature zero write $q = r^2$. The exact division (3.13) gives

$$\delta = r^2\{mH_\Delta(r, m, p) + rB_\Delta(r, p)\}. \tag{4.5}$$

Here H_Δ has sign A, and B_Δ has sign c_0/A, with uniform nonzero margins. In the sign case forced by six zeros, mH_Δ has the sign of δ, whereas rB_Δ has the opposite sign. Equation (4.5) consequently implies

$$r \le K|m|, \qquad |v| = |\delta|/r^2 = |mH_\Delta + rB_\Delta| \le K|m| \tag{4.6}$$

at both zeros of P''.

The derivative Q', taken at fixed parameters, has the smooth root expression (3.4). It equals one at $r = 0$, so $Q' - 1 = rU_1(r, m, v, p)$ with U_1 smooth. The aligned strength cancellation gives $U_1(0, 0, 0, p) = 0$. Hence

$$|Q' - 1| \le Kr(r + |m| + |v|). \tag{4.7}$$

At ζ_2, equations (4.6) and (4.7), together with $Q = O(r^2)$, yield $D'(\zeta_2) = b + O(m^2)$. At the first threshold $Q' = 1$. For positive separation that threshold is zero and $Q = 0$, giving $D'(\theta_1) = b$; for negative separation it is δ, with $Q = \delta$, giving $D'(\theta_1) = b + O(|\delta|)$. The first and last inequalities in (4.3) therefore imply

$$-Km^2 \le sb \le K|\delta|.$$

The cubic separation bound proves $|b| \le K|m|^2$.

For the offset and the states, use the bound (2.21), which is uniform up to the thresholds.

On the inactive side the exact closing expansion gives

$$E(q) = sa + sbq + sc(\lambda)q^2 + O(|q|^3).$$

After shrinking the state interval, its last two terms are bounded above by $-\gamma q^2$ for one $\gamma > 0$. Completing the square gives $E(q) \le sa + Kb^2$. The positive inactive maximum from Lemma 4.2 therefore implies

$$sa \ge -Kb^2. \tag{4.8}$$

The next, negative minimum lies between the two thresholds, where $|q| \le K|\delta|$. Equation (2.21) and the closing expansion give there

$$|D(q) - a| \le K\{|b\delta| + |\delta|^{3/2} + \delta^2\}.$$

Consequently

$$sa \le K\{|b\delta| + |\delta|^{3/2} + \delta^2\} \le K|m|^{9/2}. \tag{4.9}$$

Together with (4.8), this proves $|a| \le K|m|^4$.

The three parameter bounds just proved satisfy the hypotheses of Lemma 4.1, which confines every zero to $|q| < Mm^2$. □

4.1 The four-root limit

The parameter and state weights are now forced by the six-point configuration. We next identify the coefficient geometry of its limits.

Fix the sign case $A_1(\lambda_0) = B > 0$ and $c_0 = -C < 0$, and define

$$A_4 = C^3 a/m^4, \qquad S_4 = -Cb/m^2, \qquad T_4 = 3BC^2\delta/(2m^3), \qquad x = C\sqrt{q}/m.$$

Here B and C are reference values. Write

$$\Phi_{A,S,T}(x) = A - Sx^2 + x^3 - x^4 + Tx, \qquad \mathcal{K}_4 = \{(-e_4(r), e_2(r), e_3(r)) : r_i \geq 0,\ \sum_i r_i = 1\}.$$

Let Λ be the standard unfolding, defined on a neighbourhood $\mathcal{V} \subset \mathbb{R}^4$ of 0. Write

$$N_\lambda(I) = \#\{q \in I : D_\lambda(q) = 0\}, \qquad v_m(A,S,T) = \left(\frac{m^4 A}{C^3}, -\frac{m^2 S}{C}, m, \frac{2m^3 T}{3BC^2}\right).$$

Choose a compact box $\mathcal{B}$, independent of m, containing $\mathcal{K}_4$ in its interior and the normalized parameters of every six-cycle configuration in $\mathcal{V}$; Proposition 4.3 provides such a box after shrinking $\mathcal{V}$. Define

$$\mathscr{T}_m = \{(A,S,T) \in \mathcal{B} : v_m(A,S,T) \in \mathcal{V}, N_{\Lambda(v_m(A,S,T))}(I) = 6\}. \tag{4.10}$$

For $x > 0$ put $F_m(x) = C^3 m^{-4} D(m^2 x^2/C^2)$. At fixed parameters,

$$D_q = \frac{m^2}{2Cx} F_m', \qquad D_{qq} = \frac{C}{4x^3}(xF_m'' - F_m').$$

Thus the limiting curvature equation is $x\Phi'' - \Phi' = 3x^2 - 8x^3 - T = 0$, with positive critical point $x = 1/4$, critical value $T = 1/16$, and nonzero root $x = 3/8$ when $T = 0$.

Lemma 4.4 (Convergence at the principal scale)**.** *Let parameters tend to the reference, let $m > 0$ tend to zero, and assume that (A_4, S_4, T_4) stays bounded. If these triples converge to (A,S,T), then*

$$\frac{C^3}{m^4} D(m^2 x^2/C^2) \longrightarrow \Phi_{A,S,T}(x)$$

in C^k on every compact subset of $x > 0$, for every fixed finite k.

Proof. Put $r = mx/C$ and $v = \delta/r^2 = 2mT_4/(3Bx^2)$. On a positive compact x-interval, $r, m, v \to 0$ uniformly with all fixed-order x derivatives. The division (2.20) gives

$$Q - q = mr^3 + \tfrac{3}{4}m^2 r^4 + r^5 L_0 + \delta r H, \qquad H(0,0,0,p) = \tfrac{3}{2}A_1(p,0,0).$$

Consequently $C^3(Q-q)/m^4 \to x^3 + Tx$ in every such C^k norm. In the closing expansion

$$D = a + bQ + (Q - q) + c(\lambda)Q^2 + Q^3 S_R,$$

we have $Q = q + O_{C^k}(m^4)$, $b = -m^2 S_4/C$, and $c(\lambda) \to -C$. The four remaining terms therefore converge respectively to A, $-Sx^2$, $-x^4$, and zero after multiplication by C^3/m^4. All profiles are evaluated in one compact neighbourhood, so their state derivatives are controlled however the remaining parameters converge. □

Convergence on positive compact intervals leaves open the behaviour near the thresholds. The next lemma controls the critical point that can collapse onto a threshold.

Lemma 4.5 (Critical points near the thresholds)**.** *Let parameters tend to the reference through returns with six fixed points, and let their normalized coefficients converge to* (A, S, T)*. Denote the five successive critical points of* D *by* $t_1, \dots, t_5$*, and put* $x_j = C\sqrt{t_j}/m$ *for* $j = 3, 4, 5$*. If* $T > 0$*, all three* x_j *stay in a compact subset of* $(0, \infty)$*. If* $T = 0$*, then* $x_3 \to 0$*,* $A = 0$*, and every limit of* x_5 *is at least* $3/8$*.*

Proof. The critical points interlace the six zeros, so their upper bounds follow from the state confinement in (4.4). The offset estimate (4.9) gives $a \le Km^{9/2}$, hence $A \le 0$.

Suppose first that $T > 0$ and that $x_3 \to 0$ along a subsequence. At $q = t_3$ both passages are active. Let $y = \Gamma_1(q) - \delta \ge 0$. Since $q^{3/2}/\delta$ is a fixed positive constant times x_3^3/T_4, this ratio tends to zero. More generally, on the common small parameter and state domain, assume $A_1 \ge B/2$, $m > 0$, $\delta > 0$, $q > 0$ and $y \ge 0$. If $q^{3/2}/\delta$ is sufficiently small, the passage value estimate gives

$$q - y \ge \delta/2, \quad 0 \le y < q, \quad \sqrt{q} - \sqrt{y} \ge \delta/(4\sqrt{q}).$$

The derivative expansion, with all coefficients evaluated at the same parameter, is

$$Q_q - 1 = \tfrac{3}{2}A_1(\sqrt{q} - \sqrt{y}) + \tfrac{3}{2}(A_1 + A_2)\sqrt{y} + O(q).$$

The strength identity gives $A_1 + A_2 = m > 0$. Also $Q = O(q)$, $Q_q = 1 + O(\sqrt{q})$, and $R'(Q) - 1 = b + O(q)$. Consequently, for common constants $\kappa, K > 0$,

$$D_q(q) \ge \kappa\delta/\sqrt{q} - K(|b| + q). \tag{4.11}$$

This estimate includes $y = 0$, the second threshold, by continuity of the passage derivatives. Applied at t_3 and multiplied by C/m^2, its right-hand side is bounded below by $\kappa_1 T_4/x_3 - K_1 - K_2 x_3^2$, which tends to $+\infty$. This contradicts $D_q(t_3) = 0$. The ordering $x_3 < x_4 < x_5$ proves the first assertion.

Now suppose $T = 0$. Write $r_- < r_+$ for the square roots of the two zeros of P'' in the outer region. They lie on opposite sides of the unique positive critical point r_c of the separation graph. Its exact formula (4.5) and the boundary coefficients in (3.12)–(3.13) give, for bounded $x = Cr/m$,

$$T_4 = 3x^2 - 8x^3 + o(1), \qquad Cr_c/m \longrightarrow 1/4.$$

The error is uniform down to $x = 0$ because the graph profiles are smooth. Both $Cr_\pm/m$ are bounded by the curvature-zero confinement (4.6). A limit of Cr_-/m lies in $[0, 1/4]$ and solves $x^2(3 - 8x) = 0$, hence is zero. A limit of Cr_+/m is at least $1/4$ and solves the same equation, hence is $3/8$. Since $t_3 < r_-^2$ and $t_5 > r_+^2$, the location assertions follow.

The estimate (2.21) yields

$$|D(q) - a| \le K\{|b|q + q^2 + mq^{3/2} + m^2q^2 + q^{5/2} + \delta\sqrt{q + \delta} + \delta q^{3/2}\}.$$

At $q = t_3$, division by m^4 bounds this by a constant times

$$x_3^2 + x_3^3 + x_3^4 + m^2x_3^4 + mx_3^5 + T_4x_3 + \sqrt{m}\,T_4^{3/2} + m^2T_4x_3^3,$$

which tends to zero. Since $D(t_3) > 0$, this gives $A \ge 0$, and therefore $A = 0$. □

These two convergence statements allow the critical-value signs to pass to the quartic, including its zero-root face.

Proposition 4.6 (Necessary four-root geometry)**.** *In this sign case, for every admissible family, every accumulation point of* (A_4, S_4, T_4) *along a saturated sequence belongs to* $\mathcal{K}_4$*.*

Proof. Take a convergent coefficient subsequence, using (4.4). In this sign case (4.3) gives $b = D'(0) < 0$ on every saturated return, so $S_4 = -Cb/m^2 > 0$ and its limit satisfies $S \geq 0$. If $T > 0$, Lemmas 4.4 and 4.5 give three positive limiting critical points $u \leq v \leq w$ of Φ, with

$$\Phi(u) \geq 0, \qquad \Phi(v) \leq 0, \qquad \Phi(w) \geq 0.$$

If $k \leq 3$ critical points converge to the same positive point z, repeated Rolle gives zeros of $F_m^{(j)}$ tending to z for $1 \leq j \leq k$. The C^3 convergence yields $\Phi^{(j)}(z) = 0$ for those j, so z is a zero of Φ' of multiplicity at least k. If $u < v < w$, the four successive intervals $[0, u], [u, v], [v, w], [w, \infty)$ each contain a zero, since $\Phi(0) = A \leq 0$ and Φ tends to $-\infty$. A zero at a shared critical endpoint has multiplicity at least two, so the four interval allocations are valid when counted with multiplicity. If $u = v < w$, then $\Phi(u) = 0$ and the double derivative zero gives a triple polynomial zero at u; a fourth zero lies in $[w, \infty)$. The case $u < v = w$ is analogous. If all three merge, the common value is zero and the root has multiplicity four. Thus all four roots are nonnegative and their sum is one.

If $T = 0$, Lemma 4.5 gives $A = 0$. After subselection $x_5 \to w \geq 3/8$, and positive-compact convergence gives $\Phi(w) \geq 0$. Therefore $-S + w - w^2 \geq 0$, so $0 \leq S \leq 1/4$. The factorization $\Phi = x^2(-x^2 + x - S)$ again gives four nonnegative roots with sum one. □

Completion of Theorem 1.3. The interior of $\mathcal{K}_4$ consists exactly of coefficient triples with four distinct positive roots. The set of such triples is open by simple-root continuation. A zero root or a multiple root is a boundary point: a small constant-term perturbation gives respectively a negative root or violates one of the necessary critical-value signs for four real roots. Triples with distinct positive roots are dense, by perturbing the root vector inside the simplex while preserving its sum.

For positive roots r_i of an interior triple, the construction in the proof of Proposition 3.9 with r_i/C and $\varepsilon = m$ gives exactly the parameters $v_m(A, S, T)$ and two simple negative roots. On each compact subset of the interior, uniform isolation and nonzero derivative margins let Lemma 3.8 continue all six roots in the standard unfolding for uniformly small $m > 0$; Proposition 3.4 excludes others in I.

For the Hausdorff upper inclusion, a contrary sequence of closure points can be approximated within $1/n$ by true points of $\mathscr{T}_{m_n}$. Compactness of $\mathcal{B}$ and Proposition 4.6 then force its limit into $\mathcal{K}_4$. For the lower inclusion, approximate compact $\mathcal{K}_4$ by a finite net of interior points and realize all of them at every sufficiently small m. □

4.2 Boundary geometry of the limiting domain

Figure 3 displays two sections of the limiting coefficient region; the critical-value criterion below describes their boundaries.

The symmetric mean inequalities give

$$-\frac{1}{256} \leq A_4 \leq 0, \qquad 0 \leq S_4 \leq \frac{3}{8}, \qquad 0 \leq T_4 \leq \frac{1}{16}.$$

The three coefficient extrema $A_4 = -1/256$, $S_4 = 3/8$, and $T_4 = 1/16$ are attained at $r_i = 1/4$, where $\Phi = -(x - 1/4)^4$. At the opposite face, $e_3 = 0$ forces at least two roots to vanish, and therefore

$$\mathcal{K}_4 \cap \{T_4 = 0\} = \{(0, S_4, 0) : 0 \leq S_4 \leq 1/4\}.$$

These bounds are sharp for $\mathcal{K}_4$ and hold asymptotically along saturated sequences.

For fixed S_4, T_4 with three positive critical points $u < v < w$, put $f(x) = -S_4x^2 + x^3 - x^4 + T_4x$. The admissible offset interval is

$$\max\{-f(u), -f(w)\} \leq A_4 \leq \min\{0, -f(v)\}.$$

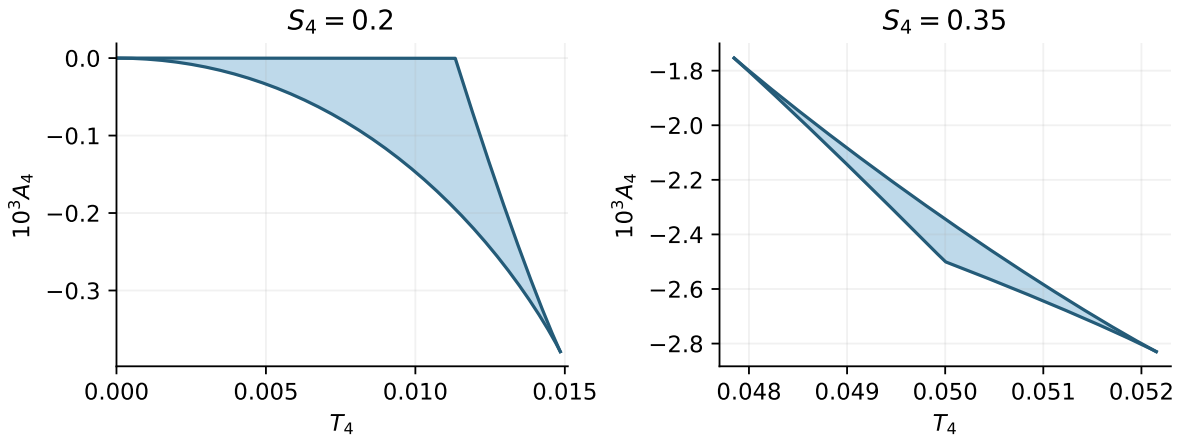


Figure 3: Sections of the limiting four-root domain at $S_4 = 0.2$ and $S_4 = 0.35$. The ordinate is $10^3 A_4$, and the two panels have different axis ranges. The boundaries are drawn from the critical-value interval described below. In the left panel the segment $A_4 = 0$ represents a zero root. The remaining boundary arcs have repeated roots; the two lower bounds switch when $f(u) = f(w)$, where the two local maxima have equal value.

It is nonempty precisely when these critical-value inequalities are compatible. Taking closures includes merged critical points in this description of the limiting boundary.

5 Limit cycles of multiplicity four and their unfolding

We continue the fourfold limiting root to a limit cycle of multiplicity four for small $m > 0$ and describe its unfolding together with the two simple fixed points with $q < 0$. We use the standard unfolding in the sign case $A_1(\lambda_0) = B > 0$, $c_0 = -C < 0$. The proof has three steps: location of all zeros in I, including near the thresholds; construction of the multiple root; and preparation of the positive cluster in a chart where D is smooth in q.

Lemma 5.1 (Values across the thresholds)**.** *Set*

$$a = \frac{m^4 A}{C^3}, \qquad b = -\frac{m^2 S}{C}, \qquad \delta = \frac{2m^3 T}{3BC^2}, \qquad q = \frac{m^2 \zeta}{C^2}, \qquad m > 0. \tag{5.1}$$

For every compact set of (A, S, T) and every bounded ζ-interval, uniformly on their product,

$$\frac{C^3}{m^4} D\left(\frac{m^2 \zeta}{C^2}\right) = A - S\zeta - \zeta^2 + \zeta_+^{3/2} + T\sqrt{\zeta_+} + O(\sqrt{m}). \tag{5.2}$$

This is a C^0 estimate, uniform across both thresholds.

Proof. Hold a, b, m fixed and integrate in the separation t from 0 to δ. Put $y = \Gamma_1(q, t) - t$. The passage jet and its parameter derivatives give

$$\Gamma_{\nu,q}(x) = 1 + \tfrac{3}{2} A_\nu \sqrt{x_+} + O(x_+), \qquad \Gamma_{\nu,t}(x) = O(x_+^{3/2}), \qquad |y - q| \le K(q_+^{3/2} + |t|).$$

These estimates include zero inputs. Along the integration segment, $A_2 = -B + O(m)$ and $|t| = O(m^3)$. The square-root inequality $|\sqrt{y_+} - \sqrt{q_+}| \le \sqrt{|y - q|}$ and the total derivative (2.23) therefore give

$$\begin{aligned}\left|Q_t - \tfrac{3}{2} B\sqrt{q_+}\right| \le K\{&q_+^{3/4} + \sqrt{|t|} + m\sqrt{q_+} + q_+ + |t| \\ &+ q_+^{3/2} + (q_+ + |t|)^{3/2}\} = O(m^{3/2}).\end{aligned}$$

This bound uses $|t|$ and remains valid for $q < 0$, so both signs of separation are included. Integration and the aligned identity (2.13) yield

$$Q - q = mq_+^{3/2} + \tfrac{3}{2}B\delta\sqrt{q_+} + O(m^{9/2}), \qquad Q - q = O(m^4).$$

In (2.11), $c = -C + O(m)$ because the remaining parameters are fixed at ξ_0. Replacing Q by q in $bQ + cQ^2$ changes these terms by $O(m^6)$, while $(c + C)q^2 = O(m^5)$ and $Q^3S = O(m^6)$. Thus

$$D = a + bq - Cq^2 + mq_+^{3/2} + \tfrac{3}{2}B\delta\sqrt{q_+} + O(m^{9/2}).$$

Substitution of (5.1) proves the assertion. □

With (5.1), set $\Theta = (A, S, T)$ and

$$\Theta_0 = \left(-\frac{1}{256}, \frac{3}{8}, \frac{1}{16}\right), \qquad x_0 = \frac{1}{4}. \tag{5.3}$$

To count cycles on both sides of this bifurcation, we first locate all zeros near the fourfold configuration.

Lemma 5.2 (All roots near the quadruple configuration)**.** *There are a neighbourhood $\mathcal{O}$ of Θ_0, $m_0 > 0$ and three disjoint compact state intervals $J_{N,-}, J_{N,+}, J_Q$ about, respectively,*

$$\zeta_{N,\pm} = \frac{-3 \pm 2\sqrt{2}}{16} < 0, \qquad \zeta_Q = \frac{1}{16} > 0,$$

such that, for every $\Theta \in \mathcal{O}$ and $0 < m < m_0$, all zeros in I lie in $(m^2/C^2)(J_{N,-} \cup J_{N,+} \cup J_Q)$. Each negative interval contains exactly one simple zero. Both passages are active on the positive interval, which contains all remaining zeros. The three intervals can be chosen arbitrarily small about their respective centres, by reducing $\mathcal{O}$ and m_0.

Proof. Bounded parameters satisfy the hypotheses of Lemma 4.1. Thus all zeros have $|\zeta| \le L$ for a fixed L. At Θ_0, the full-state limit in (5.2) is

$$-(\zeta - \zeta_{N,-})(\zeta - \zeta_{N,+}) \quad (\zeta < 0), \qquad -\left(\sqrt{\zeta} - \tfrac{1}{4}\right)^4 \quad (\zeta \ge 0).$$

Its only root positions are the three stated centres, and its value at zero is $-1/256$. Choose the three small intervals with each centre in its interval's interior. The complement of their interiors in $[-L, L]$ is compact and contains no limiting root, so the absolute value of the limiting function has a positive minimum there. First reducing $\mathcal{O}$, then m_0, preserves that gap by Lemma 5.1. This excludes all zeros outside the interval interiors, including their boundaries and the threshold layer.

Take $\mathcal{O}$ so that $T > 0$. For $q < 0$ both passages are inactive, so the closing expansion gives on either negative interval

$$\frac{C^3}{m^4}D(m^2\zeta/C^2) = A - S\zeta - \zeta^2 + O_{C^1}(m).$$

The limiting derivatives at $\zeta_{N,-}, \zeta_{N,+}$ are $\sqrt{2}/4, -\sqrt{2}/4$. Their strict signs and the opposite endpoint values persist uniformly. The intermediate value theorem and strict monotonicity give one simple root in each interval. In particular, their scaled locations are

$$\frac{-S \pm \sqrt{S^2 + 4A}}{2} + O(m).$$

Finally the second threshold is $a_\delta = \Gamma_1^{-1}(\delta) = O(m^3)$, whereas the positive interval has $q \ge \kappa m^2$ for a fixed $\kappa > 0$. Hence that entire interval lies beyond both thresholds. □

The localization fixes the two negative zeros and places the positive cluster beyond both thresholds. It remains to meet the four zero conditions simultaneously and identify the local unfolding. The following theorem refines Theorem 1.4.

Theorem 5.3 (Limit cycles of multiplicity four)**.** *After a reduction of $\mathcal{O}$ and m_0 in Lemma 5.2, there is a unique smooth branch $m \mapsto (\Theta_*(m), x_*(m))$, $0 \le m < m_0$, equal to (Θ_0, x_0) at $m = 0$, such that for every $0 < m < m_0$ the displacement has a zero of multiplicity four at $q_*(m) = m^2 x_*(m)^2/C^2$ for the parameters $\Theta_*(m)$. Its parameters and state satisfy*

$$\begin{aligned} a_*(m) &= -\frac{m^4}{256C^3} + O(m^5), \quad b_*(m) = -\frac{3m^2}{8C} + O(m^3), \\ \delta_*(m) &= \frac{m^3}{24BC^2} + O(m^4), \quad q_*(m) = \frac{m^2}{16C^2} + O(m^3). \end{aligned} \tag{5.4}$$

Set $p_*(m) = (a_*(m), b_*(m), \xi_0)$. Then

$$q_*(m) = r_{\mathrm{c}}(m, p_*(m))^2, \qquad \delta_*(m) = \delta_{\mathrm{c}}(m, p_*(m)). \tag{5.5}$$

For $0 < m < m_0$ there are smooth coordinates (u, v, w) on the parameters (a, b, δ), centred on this branch, and a smooth state coordinate y, in which the local zero equation is, up to a nonvanishing smooth factor,

$$\mathfrak{p}(y) = y^4 + uy^2 + vy + w = 0. \tag{5.6}$$

In the normalized variables (m, x, A, S, T), the changes and the nonzero factor extend jointly smoothly through $m = 0$. Off the multiple-root set, the number of fixed points in I is two, four or six; each count occurs on a nonempty open set of parameters. At the fourfold point there are exactly three fixed points in I: the fourfold one and two simple ones with $q < 0$.

By (5.5), the fourfold zero lies at the positive critical point of the separation graph, where $D'' = D''' = 0$ by Proposition 3.7; the conditions $D = D' = 0$ are additional. The normalized fourfold-root equations construct the branch, and preparation then gives three independent unfolding parameters on the positive interval isolated above.

Proof. Lemma 3.8, after constant rescaling, gives the jointly smooth auxiliary function

$$F(m, x; \Theta) = \frac{C^3}{m^4} D(m^2 x^2/C^2) = A - Sx^2 + x^3 - x^4 + Tx + mE_4(m, x, \Theta) \tag{5.7}$$

near $(0, x_0, \Theta_0)$, with all prescribed finite mixed derivatives bounded. It equals the rescaled displacement for $m > 0$. For $H = (F, F_x, F_{xx}, F_{xxx})$, differentiation at that point, in columns (A, S, T, x), gives

$$D_{(A,S,T,x)} H = \begin{pmatrix} 1 & -1/16 & 1/4 & 0 \\ 0 & -1/2 & 1 & 0 \\ 0 & -2 & 0 & 0 \\ 0 & 0 & 0 & -24 \end{pmatrix}, \qquad \det D_{(A,S,T,x)} H = -48.$$

The implicit function theorem gives the unique local branch $\Theta_* = \Theta_0 + O(m)$, $x_* = x_0 + O(m)$, with $F_{xxxx} = -24 + O(m) \neq 0$. Substituting the weights gives (5.4). Lemma 5.2 places this zero where both passages are active. For each fixed $m > 0$, $q = m^2 x^2/C^2$ is a local diffeomorphism there, so its multiplicity in q is four.

Put $r_* = \sqrt{q_*}$ and $p_* = (a_*, b_*, \xi_0)$. By (5.4),

$$r_* = \frac{m}{4C} + O(m^2), \qquad \frac{\delta_*}{q_*} = \frac{2m}{3B} + O(m^2).$$

After reducing m_0, this point lies in the outer region of (3.17). At fixed (m, p, δ), set $K(r) = r^{-3}\mathcal{U}(r, m, \delta/r^2, p) \neq 0$. That factorization and differentiation at fixed parameters give

$$D''(r^2) = K(r)[\delta - \Delta(r, m, p)],$$
$$2rD'''(r^2) = K'(r)[\delta - \Delta(r, m, p)] - K(r)\Delta_r(r, m, p).$$

At the quadruple point $D'' = D''' = 0$, hence $\delta_* = \Delta(r_*, m, p_*)$ and $\Delta_r(r_*, m, p_*) = 0$. Since $r_* > 0$ and $c_0(p_*) < 0$, Proposition 3.7 identifies this critical point as $r_* = r_c(m, p_*)$, proving (5.5).

The Malgrange preparation theorem [10, Chapter IV, Section 2, Theorem 2.1 and Note (1), p. 95] in $t = x - x_0$, with parameters (m, Θ), followed by a translation removing the cubic term, gives (5.6). At $m = 0$, $\Theta = \Theta_0$, the unit is -1 and the lower coefficients vanish. Writing $\Delta\Theta = \Theta - \Theta_0$, comparison modulo t^3 at $m = 0$ gives

$$u = \Delta S + O(|\Delta\Theta|^2),$$
$$v = 2x_0\Delta S - \Delta T + O(|\Delta\Theta|^2),$$
$$w = -\Delta A + x_0^2\Delta S - x_0\Delta T + O(|\Delta\Theta|^2).$$

The determinant in $(\Delta A, \Delta S, \Delta T)$ is 1; equivalently, $\det D_{(A,S,T)}(F, F_x, F_{xx}) = 2 + O(m) \neq 0$ on the branch. Thus $(m, \Theta) \mapsto (m, u, v, w)$ is a local smooth coordinate change in the normalized variables through $m = 0$, centred on the branch. For each fixed $m > 0$, (u, v, w) are local coordinates on the slice of parameters with this value of m; preparation preserves multiplicities.

Choose the positive interval of Lemma 5.2 inside this preparation chart. A common reduction of the parameters confines every real quartic root there, since the coefficients vanish at the centre. The whole positive cluster therefore has 0, 2 or 4 simple roots off the multiple-root set. The polynomials

$$y^4 + \eta^4, \qquad y^4 - \eta^4, \qquad (y^2 - \eta^2)(y^2 - 4\eta^2), \qquad 0 < \eta \ll 1,$$

realize all three counts on open coefficient sets. The parameter diffeomorphism and the two simple negative roots give the stated counts, including three fixed points at the origin. □

Proposition 5.4 (Local quartic strata)**.** *In the coordinates (u, v, w) of (5.6), the multiple-root set is parametrized by*

$$v = -4t^3 - 2ut, \qquad w = 3t^4 + ut^2. \tag{5.8}$$

Its triple-root curve, its two-double-root curve and its quadruple point are respectively

$$(u, v, w) = (-6t^2, 8t^3, -3t^4), \quad t \neq 0,$$
$$(u, v, w) = (-2t^2, 0, t^4), \qquad t \neq 0,$$
$$(u, v, w) = (0, 0, 0).$$

The ordinary fold part of (5.8) excludes both the triple-root curve and a simultaneous second double root.

Proof. Solving $\mathfrak{p}(t) = \mathfrak{p}'(t) = 0$ gives (5.8). On this surface, $\mathfrak{p}''(t) = 0$ gives $\mathfrak{p} = (y - t)^3(y + 3t)$, and the additional equation $\mathfrak{p}'''(t) = 0$ forces $t = 0$. Two distinct double roots must be $t, -t$, since the cubic coefficient vanishes, so $\mathfrak{p} = (y^2 - t^2)^2$. □

Figure 4 displays the local root counts in these coordinates; the coordinate change depends on m. These strata describe a neighbourhood of the fourfold cycle at each fixed small $m > 0$.

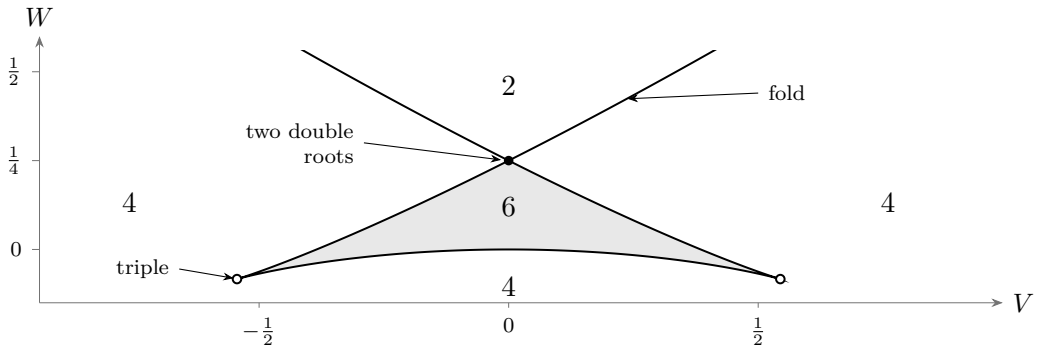


Figure 4: A slice of the prepared quartic $y^4 + uy^2 + vy + w$ at $u = -\varepsilon^2$, for any sufficiently small $\varepsilon > 0$. In the similarity coordinates $V = v/\varepsilon^3$, $W = w/\varepsilon^4$, the multiple-root set is $(V, W) = (2t - 4t^3, 3t^4 - t^2)$. Region labels are the total numbers of simple cycles, including the two simple fixed points with $q < 0$. Open circles mark triple roots; the crossing has two double roots. This slice does not contain the quadruple-root point $u = v = w = 0$.

6 Realization in fields and a harvesting model

Two constructions show that the hypotheses and the rank condition (2.12) hold. Local perturbations of an arbitrary reference field give four independent unfolding parameters, and a two-tier harvesting model gives them through its natural parameters. In both cases we verify the first-return geometry and use the standard unfolding to transfer the cycle configurations.

6.1 Local perturbations of a given reference field

Fix a nonstationary reference cycle $\mathcal{C}$ satisfying the geometric hypotheses of Section 2, with passage strengths $A_{1,0} = -A_{2,0} \neq 0$ and closing jets

$$R_0(0) = 0, \qquad R_0'(0) = 1, \qquad c_0 = \tfrac{1}{2}R_0''(0) \neq 0.$$

The positive penetration directions agree under the regular connections, which have no additional seam crossing. Choose a second-contact box U_2 and a smooth closing-arc flow box U_3, with disjoint closures away from the first contact and return section. Let $\mathcal{L}_{\mathcal{C}}(U_2, U_3)$ be the finite-dimensional jointly smooth parameter germs through this field with seam changes supported in $\operatorname{int} U_2$, inactive branch fixed there, and active difference $h_\lambda Z_\lambda$ with smooth Z_λ. In $\operatorname{int} U_3$, which stays away from the seam, a common smooth field may be added to both branches. All other data are fixed. The perturbations that control (m, δ) and (a, b) have disjoint supports.

Proposition 6.1 (Four independent unfolding parameters)**.** *Every such reference field admits a four-parameter unfolding in $\mathcal{L}_{\mathcal{C}}(U_2, U_3)$ with $D_\lambda(a, b, m, \delta)$ invertible at $\lambda = 0$, which reduces to the reference field at $\lambda = 0$. The perturbations preserve matching, regular seams, quadratic contacts and a one-turn first-return interval. Every finite-parameter germ in $\mathcal{L}_{\mathcal{C}}(U_2, U_3)$ is an admissible family after a reduction depending on the germ.*

Proof. Transport the first inactive integral along the unchanged first connector. A Morse-type normalization of the clock near the second contact gives

$$X_0^- = v(t, q)\partial_t, \qquad X_0^+ = v(t, q)\{[1 + (q - t^2)\alpha(t, q)]\partial_t + (q - t^2)\mu(t, q)\partial_q\}, \quad v > 0.$$

For a positive smooth e, $h_0 = e(q - t^2)$ and $eZ_0 = v(\alpha\partial_t + \mu\partial_q)$. Choose compactly supported cutoffs ψ_1, ψ_2 in U_2, equal to one on a smaller contact core, and set

$$h_d = h_0 - de\psi_1, \qquad X_{\kappa,d}^+ = X_0^+ + (h_d - h_0)Z_0 + \kappa h_d v\psi_2 e^{-1}\partial_q, \qquad X_{\kappa,d}^- = X_0^-. \tag{6.1}$$

These changes extend by zero, recover $X_0^{\pm}$ at zero, and satisfy

$$X_{\kappa,d}^{+} - X_0^{-} = h_d(Z_0 + \kappa v \psi_2 e^{-1}\partial_q).$$

Thus the two fields agree on the moving seam. In the core, $z = q - d$ gives the normalized seam $z = t^2$ and coefficients $v(t, z+d), \alpha(t, z+d), \mu(t, z+d) + \kappa$. The contact is $(0, d)$, with inactive second seam derivative $-2v(0,d)^2 < 0$ in the normalized defining function, and the active clock stays positive. Proposition 2.4 therefore gives $A_2(\kappa, d) = \frac{4}{3}[\mu(0,d) + \kappa]$. The first passage and connector are unchanged, so

$$\delta = d, \quad m = A_{1,0} + \tfrac{4}{3}[\mu(0,d) + \kappa], \qquad D_{(\kappa,d)}(m,\delta)(0) = \begin{pmatrix} 4/3 & (4/3)\mu_q(0,0) \\ 0 & 1 \end{pmatrix}. \tag{6.2}$$

The second strength is already transported; its incoming threshold is $\Gamma_1^{-1}(d)$.

In U_3, use the base flow time s and the second out-port coordinate transported by that flow as z. Then $X_0^{-} = \partial_s$, the entrance holonomy is $q \mapsto z = q$, and the remaining transition is R_0. Add to both branches

$$Y_\eta = \chi_1(s)\chi_2(z)(\eta_0 + \eta_1 z)\partial_z, \qquad \chi_1 \geq 0, \qquad \int \chi_1(s)\, ds = 1,$$

with compact support and $\chi_2 = 1$ on a fixed transverse interval. Small inputs and parameters stay in that interval, so the box holonomy and closing transition are

$$H_\eta(z) = e^{\eta_1} z + \eta_0 \phi(\eta_1), \qquad \phi(w) = \int_0^1 e^{wt}\, dt, \qquad R_\eta = R_0 \circ H_\eta.$$

Consequently

$$D_{(\eta_0,\eta_1,\kappa,d)}(a,b,m,\delta)(0) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 2c_0 & 1 & 0 & 0 \\ 0 & 0 & 4/3 & (4/3)\mu_q(0,0) \\ 0 & 0 & 0 & 1 \end{pmatrix}, \qquad \det = 4/3. \tag{6.3}$$

Disjoint supports make the closing leg independent of (κ, d). The inverse function theorem gives the local coordinates (a, b, m, δ).

The seam $q - t^2 - d\psi_1(t,q) = 0$ has q derivative greater than $1/2$ for small d and agrees with the old seam on a boundary collar. The reference has a positive angular clock on a fixed tube; matching gives common local Lipschitz bounds for the perturbed fields. Choose cores where the cutoffs are one, ports with strict transverse margins, and strictly inactive connecting arcs, including the cutoff collars. Proposition 2.4 gives the first entry and the first exit in each core, with second penetration $\Gamma_1(q) - d$. Lemma 2.2 therefore gives the one-turn return

$$P_\lambda(q) = R_\eta\{d + \Gamma_{2,\kappa,d}(\Gamma_1(q) - d)\}.$$

For any finite-parameter germ in $\mathcal{L}_{\mathcal{C}}(U_2, U_3)$, matching, smooth parameter dependence and the implicit function theorem at the contact preserve these clock, core, port and positive transverse-derivative margins after a reduction depending on the germ. □

Corollary 6.2 (Cyclicity in the local perturbation class)**.** *For every reference cycle $\mathcal{C}$ of Proposition 6.1,* $\mathrm{Cycl}(\mathcal{L}_{\mathcal{C}}(U_2, U_3), \mathcal{C}) = 6$ *in the sense of Definition 1.1. The corresponding suprema over germs contained in $\{m = 0\}$, in $\{\delta = 0\}$ and in $\{m = \delta = 0\}$ are four, four and two. The same values hold for the class of all admissible families through the reference field.*

Proof. Each germ is admissible after its own reduction, so Proposition 3.4 gives at most six fixed points in I; the same holds for every admissible family. Remark 2.3 identifies the isolated fixed points in the common interval with the nearby limit cycles. Conversely, Proposition 6.1 and Proposition 3.9 give six simple cycles approaching $\mathcal{C}$. Restricting the same unfolding to the three slices, Proposition 3.10 attains the bounds of Corollary 3.5. Since the families of Proposition 6.1 are admissible, the same values hold for the class of all admissible families; this proves part 3 of Theorem 1.2. □

6.2 A two-tier escapement-harvesting model

This model shows that the hypotheses and the rank condition hold for the natural parameters of an ecological family. Continuous threshold harvesting, which switches harvest on continuously above a population threshold, is studied in [16, 29]. The policy of Bohn, Rebaza and Speer [1] stops prey harvest below a lower threshold, increases it linearly between two thresholds and caps it above the upper one; Onana Ndzana et al. [27] use it in a delayed Leslie–Gower model with prey refuge. We intensify harvest above the upper threshold, using the normalized smooth predator-harvesting core of Huang, Gong and Ruan [12, equation (9) and Theorem 3.6]:

$$\dot{x} = x(1-x-y) - u(x-L)_+ - v(x-U)_+, \qquad \dot{y} = y\left(d - \beta\frac{y}{x}\right) - h. \tag{6.4}$$

Here $x, y > 0$ are prey and predator densities, $\beta, d, h, u, v > 0$, and $0 < L < U < 1$. The predator parameters d, β, h describe growth, density limitation relative to prey abundance, and constant-yield harvest. The marginal prey efforts are $0, u, u+v$. We work in a compact subset of the positive quadrant. The middle smooth extension is

$$F_c = x(1-x-y) - u(x-L), \qquad G = y(d-\beta y/x) - h, \qquad X_c = (F_c, G), \tag{6.5}$$

and the exterior extensions are

$$X_\ell = (x(1-x-y), G), \qquad X_r = (F_c - v(x-U), G).$$

The adjacent extensions agree on the whole seams: $X_\ell = X_c$ on $x = L$ and $X_c = X_r$ on $x = U$. The middle extension is inactive at both contacts, with active sides outside $L \le x \le U$. The branches are analytic for $x > 0$ and the field is locally Lipschitz.

Theorem 6.3 (Natural harvesting realization)**.** *There exists $\rho_0 > 0$ such that, for every fixed radius $0 < \rho < \rho_0$ in the smooth-core fold construction of Appendix A, there exists $u_0(\rho) > 0$ with the following property. For every fixed $0 < u < u_0(\rho)$ there are positive parameters β, d, h, v and thresholds $0 < L < U < 1$, depending on ρ and u, for which (6.4) has an embedded semistable periodic orbit $\mathcal{C}$ with exactly two quadratic contacts, at its minimum and maximum prey densities. In one transported return coordinate the contact levels agree and*

$$A_1 = A_0 = -A_2 \neq 0, \qquad R(0) = 0, \qquad R'(0) = 1, \qquad c_0 = \tfrac{1}{2}R''(0) < 0.$$

The map $\lambda \mapsto (a, b, m, \delta)$ has rank four in the natural family $\lambda = (\beta, d, h, u, v, L, U)$. The cyclicity of $\mathcal{C}$ in this family is six, and the corresponding values on $m = 0$, on $\delta = 0$ and on their intersection are four, four and two. The conclusions of Theorems 1.3 and 1.4 hold for the standard unfolding of this family (Corollary 6.4).

We construct a smooth cyclic fold, fit the thresholds to its prey extrema, and cancel the passage strengths with a positive effort ratio. The directions (ν, L, v, U), with β, u fixed, then give the standard unfolding.

6.3 Threshold fitting, cancellation and closing curvature

Appendix A constructs cyclic folds of the smooth core, for every sufficiently small fixed radius $\rho > 0$. The fold is embedded and nonstationary, lies in the positive quadrant, and has exactly one nondegenerate prey minimum and maximum. In radial coordinates its displacement satisfies (A.8), with $\ell < 0$, and the (β, ν) Jacobian of (D, D_r) is invertible. These strict properties allow the thresholds and positive efforts to be fitted to a fixed fold.

For positive u, use

$$y_e = \tfrac{4}{5} - u(1 - 5L), \qquad d = 8\beta + \tfrac{1}{5} + \nu, \qquad h = y_e(d - 5\beta y_e). \tag{6.6}$$

The middle field vanishes at $(1/5, y_e)$. At $u = 0$ the chart reduces to (A.1) and is independent of L. Fix a positive fold radius in (A.8). The invertible (β, ν) Jacobian of (D, D_r) continues a fold $\mathcal{C}(u, L)$ for small $u, L - L_0$, where L_0 is the unperturbed prey minimum. Its curvature and two nondegenerate extrema persist. For its minimum $x_{\min}(u, L)$, $\partial_L x_{\min}(0, L_0) = 0$, so

$$L = x_{\min}(u, L), \qquad U = \max_x \mathcal{C}(u, L(u)) \tag{6.7}$$

has a unique local smooth solution with $0 < L(u) < U(u) < 1$. After reducing u, all parameters in (6.6) remain positive. The fitted orbit stays in the closed middle strip, touching its boundary only at the extrema, and is an orbit of (6.4) for every v. There $F_c = 0$, $F_{c,y} = -x$, and nonstationarity gives $G \neq 0$, so

$$\ddot{x} = F_{c,y} G = -xG \neq 0. \tag{6.8}$$

Figure 5 illustrates this fitted threshold geometry. For each of two fixed parameter tuples, Online Resource 3 proves by interval arithmetic the existence of at least six distinct simple periodic orbits of (6.4). Theorem 6.3 is analytic and does not depend on these certificates.

Positive penetration is outward, into $x < L$ or $x > U$. At $p_i = (x_i, y_i)$, write $G_i = G(p_i)$. Equation (6.8) gives $G_1 < 0 < G_2$, so $y = y_i$ are transverse sections. In the coordinates $z_1 = L - x$, $z_2 = x - U$, their inactive transitions along the forward arcs satisfy

$$S_1'(0) = -\frac{G_1}{G_2} \exp \int_{p_1}^{p_2} \operatorname{div} X_c \, dt > 0, \qquad S_2'(0) = -\frac{G_2}{G_1} \exp \int_{p_2}^{p_1} \operatorname{div} X_c \, dt > 0.$$

Positive penetration normalizations preserve these signs and, on a common smaller domain, give positive derivative margins. Transport one inactive first integral I from the first box through the connector to the second, with outward direction positive. These are local first-integral charts joined along the connecting tube. At p_i, put $\sigma_1 = -1$, $\sigma_2 = 1$, $s_i = I_x(p_i)$. Then $I_y(p_i) = 0$ and $\operatorname{sgn} s_i = \sigma_i$. A Morse-type normalization of the clock for $\mathfrak{h}_i = \sigma_i(x - x_i)$, preserving I, gives

$$\mathfrak{h}_i = e_i(t, I)(I - t^2), \qquad e_i(0, 0) = \frac{\sigma_i}{s_i} > 0, \qquad \dot{t}(0, 0) = \omega_i = \sqrt{-s_i F_{c,y}(p_i) G(p_i)/2} > 0. \tag{6.9}$$

These clocks differ from the core frequency ω. For active difference $\mathfrak{h}_i Z_i$, the normalized transverse coefficient is

$$\mu_i(0, 0) = \frac{e_i(0, 0)}{\omega_i} dI(Z_i) = \frac{\sigma_i (Z_i)_x}{\omega_i}.$$

Here $Z_1 = -u\partial_x$, $Z_2 = -v\partial_x$; thus (2.6) gives

$$A_1 = K_1 u = \frac{4u}{3\omega_1} > 0, \qquad A_2 = -K_2 v = -\frac{4v}{3\omega_2} < 0. \tag{6.10}$$

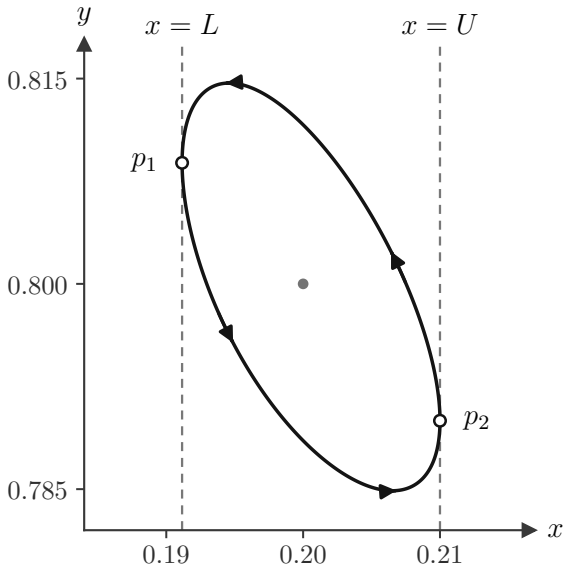


Figure 5: Numerical illustration of the fitted threshold geometry of (6.4). With $u = 10^{-6}$ and $U = 0.21$ prescribed, the resulting parameter values are $\beta \simeq 0.031443254$, $d \simeq 0.451546877$, $h \simeq 0.260619080$, and $L \simeq 0.191154841$. The middle orbit touches the vertical thresholds $x = L$ and $x = U$ at its prey extrema. Open circles mark the contacts p_1, p_2, arrows show increasing time, and the dot marks the middle equilibrium. Theorem 6.3 proves the existence of such reference cycles and their six-cycle unfoldings.

The opposing passage signs come from the opposite outward directions at the two thresholds; both harvesting efforts are positive. The inactive clocks and transport are independent of v, so the exact positive ratio

$$v = u\frac{K_1}{K_2} = u\frac{\omega_2}{\omega_1} \tag{6.11}$$

cancels the strengths in the same transported coordinate. Both contacts lie on one inactive orbit, so their levels agree. Their passage derivatives equal one at the touch and the middle fold has multiplier one. Lemma 2.5 gives $Q(q) = q + q_+^{5/2}\mathcal{L}(\sqrt{q_+})$, with no terms of order $q^{3/2}$ or q^2. The regular map R is the inactive monodromy.

The closing curvature c_0 is negative in the outward-positive coordinate. First set $u = 0$ and fix the cyclic-fold parameters at a small radius ρ. Vary the initial radial coordinate r with these parameters fixed. Smooth finite-time normal-form flow has prey component

$$x(r, \theta) = \tfrac{1}{5} + rp(\theta) + O(r^2 + \rho^2 r),$$

where p is a nonzero first harmonic and the r-derivative of the remainder is $O(r+\rho^2)$. Its unique nearby prey minimum $M(r)$ satisfies $M'(\rho) = -\gamma + O(\rho) < 0$, $\gamma > 0$. The minimum locus $F_c = 0$ is transverse, since its derivative along the orbit is $\ddot{x} \neq 0$. If $H(r)$ is the transported inactive integral there, $I_y = 0$, $I_x < 0$ give $H'(\rho) = I_x M'(\rho) > 0$. For the radial core return P_c at these fixed parameters, $R = H \circ P_c \circ H^{-1}$. At the fixed point, where $R' = 1$,

$$R''(0) = \frac{P_c''(\rho)}{H'(\rho)} < 0$$

by (A.8) and $\ell < 0$. Continuation to fitted small positive u preserves this strict sign and

$H' > 0$. The cancelling composition has zero quadratic defect, so $c_0 < 0$ for the reference cycle as well.

6.4 Rank condition and first return

At the fitted orbit we now show that the four return-map controls can be varied independently and that nearby orbits keep the first-return structure.

Use (ν, L, v, U) in (6.6), with β, u fixed. Choose the first incoming coordinate from the middle extension and lower seam only. The regular monodromy R is independent of v, U: changes of the upper ports alter the intermediate inactive holonomies but preserve their middle-return composition. The threshold separation δ is independent of v. At the upper tangency $F_c(U, y_2(U)) = 0$ and $I_y = 0$, so

$$\delta_U = I_x(p_2) \neq 0, \qquad m_v = -K_2 \neq 0. \tag{6.12}$$

Hence

$$D_{(\nu,L,v,U)}(a, b, m, \delta) = \begin{pmatrix} a_\nu & a_L & 0 & 0 \\ b_\nu & b_L & 0 & 0 \\ * & * & -K_2 & * \\ * & * & 0 & \delta_U \end{pmatrix}. \tag{6.13}$$

For the upper-left minor first set $u = 0$ at the fixed core fold. The lower tangency is $p_1(L) = (L, 1 - L)$; its level in an unchanged core transverse coordinate has derivative

$$\zeta_L = dI(p_1'(L)) = I_x(p_1) \neq 0. \tag{6.14}$$

Choose this coordinate to agree with the incoming one at the base, and centre the contact by $T_L(\zeta(L)) = 0$. Then $R_L = T_L \circ P_c \circ T_L^{-1}$. Since at the fold $P_c(\zeta) = \zeta$ and $P_c'(\zeta) = 1$,

$$a_L = 0, \qquad b_L = P_c''(\zeta)\zeta_L = 2c_0\zeta_L \neq 0. \tag{6.15}$$

At $u = 0$, the core return P_c is independent of L, so the L-dependence of R_L comes only from the moving section. Because the base coordinates agree and $D_r = 0$, contact motion leaves the offset derivative in ν unchanged; (A.7) gives $a_\nu \neq 0$. Moving-coordinate terms in b_ν do not enter the minor since $a_L = 0$. Thus the minor is $a_\nu\, 2c_0\zeta_L \neq 0$; another transverse scaling changes it only by a nonzero factor. These objects extend smoothly to $u = 0$, so at each fixed radius the minor persists for small positive u, with fitted thresholds and (6.11). All couplings remain in (6.13), whose determinant is the minor times $(-K_2)\delta_U \neq 0$. The chart directions embed in $(\beta, d, h, u, v, L, U)$, proving (2.12) in the natural parameters.

The fitted cycle is compact, embedded and nonstationary. On a fixed tube, the vertical seams stay regular and the matched model has common local Lipschitz bounds. The positive angular clock, transverse ports, positive transverse-derivative margins and strictly inactive connectors persist under small parameter changes. In each core, (6.9) and the smooth active difference satisfy the passage hypotheses, including first entry and first exit. Lemma 2.2 gives the chronological first return on one common interval. Thus each constructed reference satisfies the geometric first-return hypotheses as well as the rank condition.

6.5 Transfer to the natural parameters

Corollary 6.4 (Main results in the constructed families)**.** *For each reference constructed above, with its standard unfolding and interval I, the bounds and attainment of Theorem 1.2 hold. In the sign case $A_1(\lambda_0) = B > 0$, $c_0 = -C < 0$, which includes every harvesting reference, the conclusions of Theorem 1.3, Theorem 5.3 and Proposition 5.4 hold for the same unfolding on I. In the harvesting family all parameters remain positive and $0 < L < U < 1$.*

Proof. The Jacobians (6.3) and (6.13), together with Lemma 2.2, give the standard unfolding and common interval. Fix this unfolding before choosing the scale: every compact set of rescaled controls lies in the domain of its inverse Λ at sufficiently small scale. In the harvesting family the inverse maps (a, b, m, δ) to (ν, L, v, U), with β, u fixed, preserving positivity and $0 < L < U < 1$. Apply Theorems 1.2 and 1.3. In the stated sign case, Lemma 5.2 localizes all roots in I: two simple negative roots and a positive cluster beyond both thresholds. Theorem 5.3 and Proposition 5.4 give its preparation and strata. Remark 2.3 identifies the isolated fixed points with cycles, completing Theorem 6.3. □

7 Concluding remarks

A cancellation that makes the reference return map C^2 need not make its unfoldings jointly C^2. The aligned cancelling slice has cyclicity two, whereas a full-rank unfolding of all four return-map controls attains six limit cycles. Several questions remain open.

1. Theorem 1.3 describes the limit of the saturation domains. Apart from the fourfold point of Theorem 1.4 and the corner $S_4 \to 0$ treated in Online Resource 1, the number of limit cycles at positive scale for parameters near boundary points of $\mathcal{K}_4$ is not determined here.

2. Parts 2 and 3 of Theorem 1.3 and Theorem 1.4 are stated in the sign case $A_1(\lambda_0) > 0$, $c_0 < 0$. The remaining sign cases are not treated here.

3. A periodic orbit tangent to two switching lines of a discontinuous system with sliding is studied in [30]. The effect of cancelling leading terms in the presence of sliding is open.

4. The contacts here are quadratic. For tangencies of higher order the corrections have other exponents, and the interaction between cancellation and moving thresholds remains to be analysed.

A Smooth-core folds and finite focus equations

Threshold fitting uses a smooth-core cyclic fold with negative return curvature and a nonsingular two-parameter fold equation. We derive these inputs from finite focus equations and control the one-turn remainder with the required state and parameter derivatives.

A.1 The core chart and its focus coefficients

At $u = 0$, use the equilibrium-preserving chart

$$d = 8\beta + \tfrac{1}{5} + \nu, \qquad h = \tfrac{16}{5}\beta + \tfrac{4}{25} + \tfrac{4}{5}\nu. \tag{A.1}$$

The equilibrium is $(1/5, 4/5)$, with

$$J = \begin{pmatrix} -1/5 & -1/5 \\ 16\beta & 1/5 + \nu \end{pmatrix}, \qquad \operatorname{tr} J = \nu, \qquad \det J = \frac{80\beta - 1 - 5\nu}{25}.$$

The multiplicity-two weak focus of [12, Theorem 3.6] occurs at

$$\beta_0 = \frac{\sqrt{51} - 4}{100}, \qquad d_0 = \frac{-3 + 2\sqrt{51}}{25}, \qquad h_0 = \frac{4(1 + \sqrt{51})}{125}, \qquad \nu = 0. \tag{A.2}$$

The finite homogeneous equations below construct $V = \mathcal{R}^2/2 + \mathrm{O}(\mathcal{R}^3)$ in rotation coordinates of radius $\mathcal{R}$, and give

$$k_4 = \ell_1(\beta) = \frac{25\beta(2000\beta^2 + 160\beta - 7)}{4(80\beta - 1)}, \qquad k_6\big|_{\beta=\beta_0} = -\frac{155}{48} - \frac{235\sqrt{51}}{576} < 0. \tag{A.3}$$

Thus $\ell_1(\beta_0) = 0$ and $\ell_1'(\beta_0) = 17/2 + 4\sqrt{51} > 0$. At the focus, $\dot{V} = k_6\mathcal{R}^6 + \mathrm{O}(\mathcal{R}^7)$, so $\sqrt{2V}$ has nonzero negative quintic drift. The angular velocity tends to $\omega_0 = \sqrt{(-21 + 4\sqrt{51})/125} > 0$; positive radial scaling and angular clock therefore give $\ell < 0$ below. The conjugate eigenvalues of J have real part $\nu/2$, and $(\nu, \beta) \mapsto (\nu/2, \ell_1(\beta))$ has rank two at $(0, \beta_0)$.

At $u = \nu = 0$, put $\omega^2 = (80\beta - 1)/25$, $X = x - 1/5$, and $Z = (x - 1/5 + y - 4/5)/(5\omega)$. Direct substitution gives

$$\dot{X} = -\omega Z + \Phi, \qquad \dot{Z} = \omega X + \Psi, \qquad \Phi = -5\omega XZ, \tag{A.4}$$

where

$$\Psi = -\frac{80\omega X^2 Z + (125\omega^2 + 5)X^2 + (6\omega - 250\omega^3)XZ + (125\omega^4 + 5\omega^2)Z^2}{16\omega(5X + 1)}.$$

Expand $(1 + 5X)^{-1}$ through the degrees needed below and denote homogeneous terms by subscripts. For $\mathcal{R}^2 = X^2 + Z^2$, seek $V = \mathcal{R}^2/2 + V_3 + \cdots + V_6$. The equations are

$$\mathcal{L}V_n = k_n\mathcal{R}^n - H_n, \qquad \mathcal{L} = \omega(-Z\partial_X + X\partial_Z), \qquad H_n = \sum_{j=2}^{n-1} (\partial_X V_j\, \Phi_{n-j+1} + \partial_Z V_j\, \Psi_{n-j+1}). \tag{A.5}$$

Here $V_2 = \mathcal{R}^2/2$. On homogeneous polynomials of odd degree n, $\mathcal{L}$ is invertible and $k_n = 0$; for even n, take the radial obstruction

$$k_n = \frac{1}{2\pi}\int_0^{2\pi} H_n(\cos\theta, \sin\theta)\, d\theta$$

and set the X^n coefficient of V_n to zero to fix the radial kernel. Solving (A.5) successively in the monomial basis for $n = 3, 4, 5, 6$ gives $k_3 = k_5 = 0$ and the coefficients (A.3). A program in Online Resource 2, also included with the source files, reconstructs the rotated field and checks every coefficient residual of these finite equations in exact arithmetic.

A.2 Normal form and cyclic folds

The focus coefficients supply the generalized Hopf unfolding. Its weighted return turns the limiting fold into a periodic orbit with the geometry required at the thresholds.

Take the finite normal form through degree seven in a parameter-dependent complex coordinate. Its nonresonant divisors stay bounded away from zero near the positive frequency ω_0; hence the transformations and their required parameter derivatives are smooth. The retained resonant degrees are one, three, five and seven. Division by the positive angular speed and normalization of one turn give smooth coefficients η_1, η_2 for the first two radial terms, with invertible $(\nu, \beta) \mapsto (\eta_1, \eta_2)$ Jacobian. The fifth-degree coefficient equals $\ell \neq 0$ at the focus. For the generalized Hopf normal form see [14]; its coefficients for ordinary and delay equations are also derived by N. A. M. Delmeire, M. M. Bosschaert and Yu. A. Kuznetsov (preprint, 2025, arXiv:2507.18469v1). Normalize the angular clock to one turn and put $r = \rho z$, $\eta_1 = \rho^4 s_1$, $\eta_2 = \rho^2 s_2$. Then

$$\frac{dz}{d\theta} = \rho^4(s_1 z + s_2 z^3 + \ell z^5) + \mathrm{O}(\rho^6), \qquad \ell < 0. \tag{A.6}$$

On a fixed positive z-interval about one, the bound holds after two z derivatives and one derivative in (s_1, s_2), including mixed derivatives. The quintic coefficient varies by $\mathrm{O}(\rho^2)$, giving $\mathrm{O}(\rho^6)$ in the scaled equation, as does the retained seventh-degree term; the smooth degree-eight field remainder gives $\mathrm{O}(\rho^7)$. Taylor remainders preserve these state bounds, and scaled parameter differentiation introduces a factor ρ^4 or ρ^2. The angular speed retains a positive uniform margin. Integration over one turn and smooth finite-time flow dependence give the smooth-core displacement

$$\rho^{-5} D(\rho z; \rho^4 s_1, \rho^2 s_2) = s_1 z + s_2 z^3 + \ell z^5 + \mathrm{o}(1) \tag{A.7}$$

in the same norms, including angle-dependent remainders. At $z = 1$, the limiting fold equations $s_1 + s_2 + \ell = 0$, $s_1 + 3s_2 + 5\ell = 0$ have parameter determinant two. Hence for every sufficiently small positive ρ the implicit function theorem gives a cyclic fold with

$$\eta_1 = \ell\rho^4 + \mathrm{o}(\rho^4), \qquad \eta_2 = -2\ell\rho^2 + \mathrm{o}(\rho^2), \qquad D_{rr}(\rho) = 8\ell\rho^3 + \mathrm{o}(\rho^3) \neq 0. \tag{A.8}$$

The (D, D_r) parameter determinant is a nonzero coordinate factor times $2\rho^3 + \mathrm{o}(\rho^3)$. Moreover, $\partial_\nu \eta_1 \neq 0$ at the generalized Hopf point and

$$\partial_\nu D(\rho) = \rho \partial_\nu \eta_1 + \mathrm{o}(\rho) \neq 0.$$

These margins are strict at each fixed positive radius. After division by ρ, the fold orbit is C^2-close to a Hopf ellipse. Its prey derivative is close to a nonzero sinusoid, with strict sign away from two simple zeros and strict monotonicity near them. Thus the orbit is embedded and nonstationary, has exactly one nondegenerate prey minimum and maximum, and lies in a compact subset of $x, y > 0$. This is the fold used to fit the lower and upper thresholds in the harvesting model.

Acknowledgments and AI tool disclosure

OpenAI's ChatGPT and Codex were used during the development and preparation of this work. These tools assisted with exploring proof strategies, developing and checking mathematical arguments, preparing verification scripts, and revising the exposition and LaTeX source. They were also used to help identify relevant literature and potential gaps in intermediate drafts. The author is responsible for the mathematical claims, references, and final presentation of the paper.

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Data and code availability. The editable manuscript source, retained figures, and finite checking scripts accompany this paper. Online Resource 1 contains the mathematical supplement on secondary saturation and cusp geometry. Online Resource 2 reproduces the finite exact focus computation described in Appendix A, and contains further independent symbolic checks with their inputs and recorded outputs, together with a floating-point diagnostic of

the harvesting model. Online Resource 3 contains computer-assisted proofs that, at each of two fixed parameter tuples, the harvesting model has at least six distinct simple periodic orbits, with exact inputs, recorded enclosures, dependencies and reproduction instructions. The analytic results are proved in the text; the computational files provide independent checks and fixed-parameter examples.

Supplementary material captions. *Online Resource 1 (ESM_1.pdf).* Secondary Saturation and Cusp Geometry at Cancelling Double Grazing: secondary-scale domains, realization with positive closing offset and the secondary cusp. *Online Resource 2 (ESM_2.zip).* Independent symbolic verification: finite focus and passage calculations, control-rank determinant identities under the stated analytic hypotheses, and quartic algebra, with the supplied inputs, recorded outputs and a floating-point harvesting diagnostic. *Online Resource 3 (ESM_3.zip).* Independent interval return certificates for at least six simple cycles at each of two fixed harvesting parameter tuples, with input validation and recorded enclosures.

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